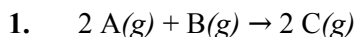


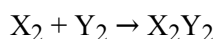
Chapter 09 Part 3: Reaction Mechanism



When the concentration of substance B in the reaction above is doubled, all other factors being held constant, it is found that the rate of the reaction remains unchanged. The most probable explanation for this observation is that

- (A) the order of the reaction with respect to substance B is 1
- (B) substance B is not involved in any of the steps in the mechanism of the reaction
- (C) substance B is not involved in the rate-determined step of the mechanism, but is involved in subsequent steps
- (D) substance B is probably a catalyst, and as such, its effect on the rate of the reaction does not depend on its concentration
- (E) the reactant with the smallest coefficient in the balanced equation generally has little or no effect on the rate of the reaction

2.



$$\text{rate} = k[\text{X}_2]$$

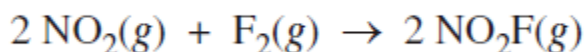
A reaction and its experimentally determined rate law are represented above. A chemist proposes two different possible mechanisms for the reaction, which are given below.

<u>Mechanism 1</u>	<u>Mechanism 2</u>
$\text{X}_2 \rightarrow 2 \text{X} \text{ (slow)}$	$\text{X}_2 \rightarrow 2 \text{X} \text{ (slow)}$
$\text{X} + \text{Y}_2 \rightarrow \text{XY}_2 \text{ (fast)}$	$\text{X} + \text{Y}_2 \rightarrow \text{XY} + \text{Y} \text{ (fast)}$
$\text{X} + \text{XY}_2 \rightarrow \text{X}_2\text{Y}_2 \text{ (fast)}$	$\text{X} + \text{XY} \rightarrow \text{X}_2\text{Y} \text{ (fast)}$
	$\text{X}_2\text{Y} + \text{Y} \rightarrow \text{X}_2\text{Y}_2 \text{ (fast)}$

Based on the information above, which of the following is true?

- (A) Only mechanism 1 is consistent with the rate law.
- (B) Only mechanism 2 is consistent with the rate law.
- (C) Both mechanism 1 and mechanism 2 are consistent with the rate law.
- (D) Neither mechanism 1 nor mechanism 2 is consistent with the rate law.

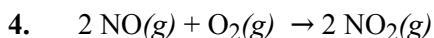
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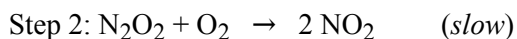
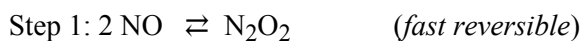
The rate law for the reaction represented by the equation above is $\text{rate} = k[\text{NO}_2][\text{F}_2]$. Which of the following could be the first elementary step of a two-step mechanism for the reaction if the first step is slow and the second step is fast?

Chapter 09 Part 3: Reaction Mechanism

- (A) $\text{F}_2(g) \rightarrow 2 \text{F}(g)$
- (B) $\text{NO}_2(g) + \text{F}_2(g) \rightarrow \text{NO}_2\text{F}(g) + \text{F}(g)$
- (C) $\text{NO}_2(g) + \text{F}(g) \rightarrow \text{NO}_2\text{F}(g)$
- (D) $2 \text{NO}_2(g) + \text{F}_2(g) \rightarrow 2 \text{NO}_2\text{F}(g)$



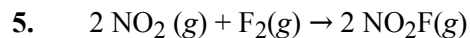
Consider the following mechanism for the reaction represented above.



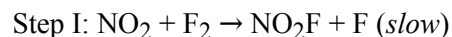
Which of the following statements is true?

- (A) Step 1 represents a unimolecular reaction.
- (B) Increasing the concentration of NO will decrease the overall rate of the reaction.
- (C) Raising the temperature will have no effect on the numerical value of the rate constant.
- (D) The rate law that is consistent with the mechanism is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$.

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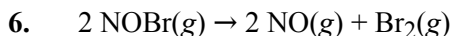


It is proposed that the reaction represented above proceeds via the mechanism represented by the two elementary steps shown below.



- Step I of the proposed mechanism involves the collision between NO_2 and F_2 molecules. This step is slow even though such collisions occur very frequently in a mixture of $\text{NO}_2(\text{g})$ and $\text{F}_2(\text{g})$. Consider a specific collision between a molecule of NO_2 and a molecule of F_2 .
 - One factor that affects whether the collision will result in a reaction is the magnitude of the collision energy. Explain.
 - Identify and explain one other factor that affects whether the collision will result in a reaction.
- Consider the following potential rate laws for the reaction. Circle the rate law below that is consistent with the mechanism proposed above. Explain the reasoning behind your choice in terms of the details of the elementary steps of the mechanism.

$$\text{rate} = k[\text{NO}_2]^2[\text{F}_2] \quad \text{rate} = k[\text{NO}_2][\text{F}_2]$$



The equation above represents an elementary step in a chemical reaction. Which of the following is the correct expression for the rate law of the elementary step?

- Rate = $k[\text{NOBr}]^{1/2}$
- Rate = $k[\text{NOBr}]$
- Rate = $k[\text{NOBr}]^2$
- Rate = $k[\text{NO}]^2 [\text{Br}_2]$



A possible mechanism for the overall reaction represented above is the following.

- $\text{NO}(\text{g}) + \text{NO}(\text{g}) \rightarrow \text{N}_2\text{O}_2(\text{g})$ *slow*
- $\text{N}_2\text{O}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$ *fast*

Which of the following rate expressions agrees best with this possible mechanism?

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(A) $\text{Rate} = k[\text{NO}]^2$

(B) $\text{Rate} = k \frac{[\text{NO}]}{[\text{O}_2]}$

(C) $\text{Rate} = k \frac{[\text{NO}]^2}{[\text{O}_2]}$

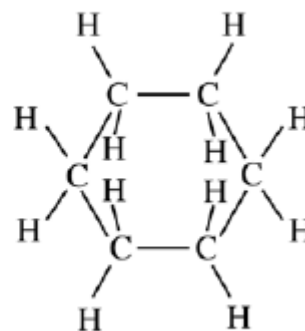
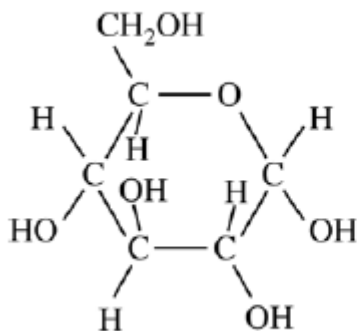
(D) $\text{Rate} = k[\text{NO}]^2 [\text{O}_2]$

(E) $\text{Rate} = k[\text{N}_2\text{O}_2] [\text{O}_2]$

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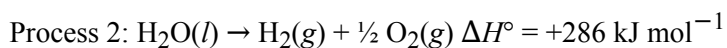
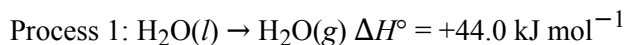
8. Answer each of the following in terms of principles of molecular behavior and chemical concepts.

- a. The structures for glucose, $C_6H_{12}O_6$, and cyclohexane, C_6H_{12} , are shown below.



Identify the type(s) of intermolecular attractive forces in

- pure glucose
 - pure cyclohexane
- b. Glucose is soluble in water but cyclohexane is not soluble in water. Explain.
- c. Consider the two processes represented below.

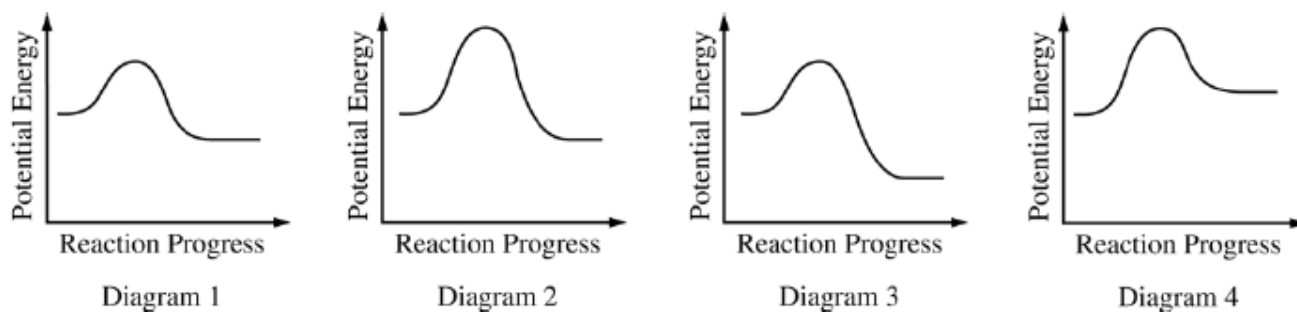


- For each of the two processes, identify the type(s) of intermolecular or intramolecular attractive forces that must be overcome for the process to occur.
- Indicate whether you agree or disagree with the statement in the box below. Support your answer with a short explanation.

When water boils, H_2O molecules break apart to form hydrogen molecules and oxygen molecules.

- d. Consider the four reaction-energy profile diagrams shown below.

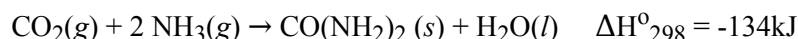
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- Identify the two diagrams that could represent a catalyzed and an uncatalyzed reaction pathway for the same reaction. Indicate which of the two diagrams represents the catalyzed reaction pathway for the reaction.
- Indicate whether you agree or disagree with the statement in the box below. Support your answer with a short explanation.

Adding a catalyst to a reaction mixture adds energy that causes the reaction to proceed more quickly.

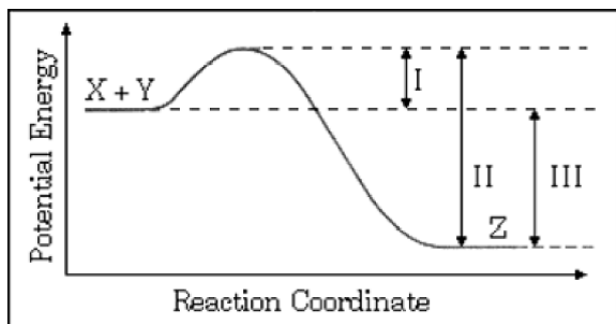
9. Answer the following questions in terms of thermodynamic principles and concepts of kinetic molecular theory.
- (a) Consider the reaction represented below, which is spontaneous at 298 K.



- For the reaction, indicate whether the standard entropy change, ΔS°_{298} is positive, or negative, or zero. Justify your answer.
 - Which factor, the change in enthalpy, ΔH°_{298} or the change in entropy, ΔS°_{298} , provides the principal driving force for the reaction at 298 K? Explain.
 - For the reaction, how is the value of the standard free energy change, ΔG° , affected by an increase in temperature? Explain.
- (b) Some reactions that are predicted by their sign of ΔG° to be spontaneous at room temperature do not proceed at a measurable rate at room temperature.
- Account for this apparent contradiction.
 - A suitable catalyst increases the rate of such a reaction. What effect does the catalyst have on ΔG° for the reaction? Explain.

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10.

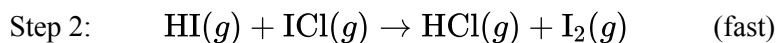
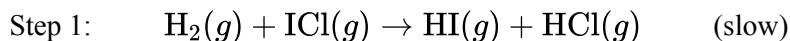


The energy diagram for the reaction $X + Y \rightarrow Z$ is shown above. The addition of a catalyst to this reaction would cause a change in which of the indicated energy differences?

- (A) I only
 - (B) II only
 - (C) III only
 - (D) I and II only
 - (E) I, II, and III
11. Step 1: $O_3(g) + Cl(g) \rightarrow O_2(g) + ClO(g)$ Step 2: $ClO(g) + O(g) \rightarrow O_2(g) + Cl(g)$

A proposed mechanism for the destruction of $O_3(g)$ in the upper atmosphere is represented above. Which of the following is the catalyst in the proposed mechanism?

- (A) $O(g)$
- (B) $Cl(g)$
- (C) $ClO(g)$
- (D) $O_2(g)$

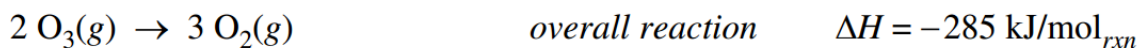
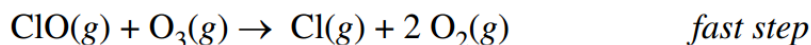
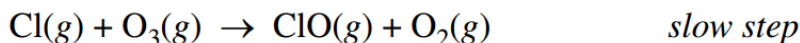


12. Which of the following represents the overall chemical equation for the reaction and the rate law for elementary step 2?

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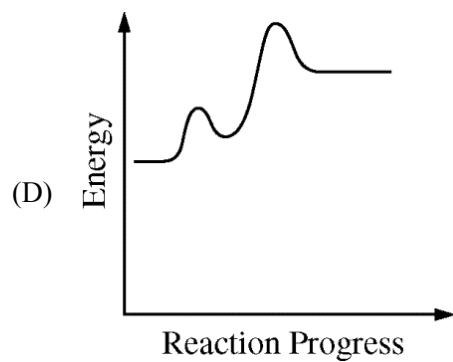
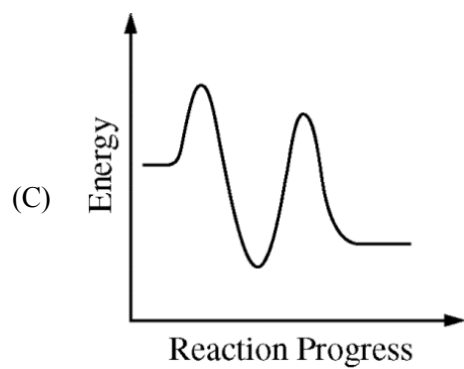
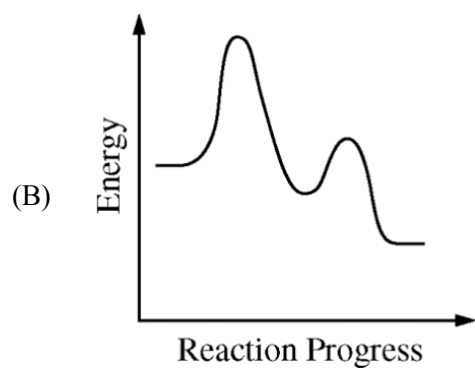
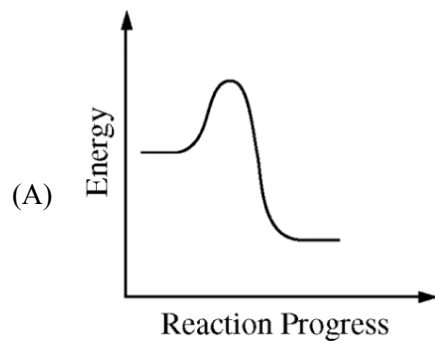
- (A) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{HI}][\text{ICl}]$.
- (B) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{ICl}]^2$.
- (C) The overall reaction is $\text{H}_2(g) + \text{ICl}(g) \rightarrow \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{ICl}]$.
- (D) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{HI}][\text{ICl}]$.
13. Which of the following represents a rate law for the overall reaction that is consistent with the proposed mechanism?
- (A) $\text{Rate} = k[\text{H}_2][\text{ICl}]$
- (B) $\text{Rate} = k[\text{H}_2][\text{ICl}]^2$
- (C) $\text{Rate} = k[\text{H}_2][\text{HI}][\text{ICl}]^2$
- (D) $\text{Rate} = k[\text{HI}][\text{ICl}]$

When free $\text{Cl}(g)$ atoms encounter $\text{O}_3(g)$ molecules in the upper atmosphere, the following reaction mechanism is proposed to occur.

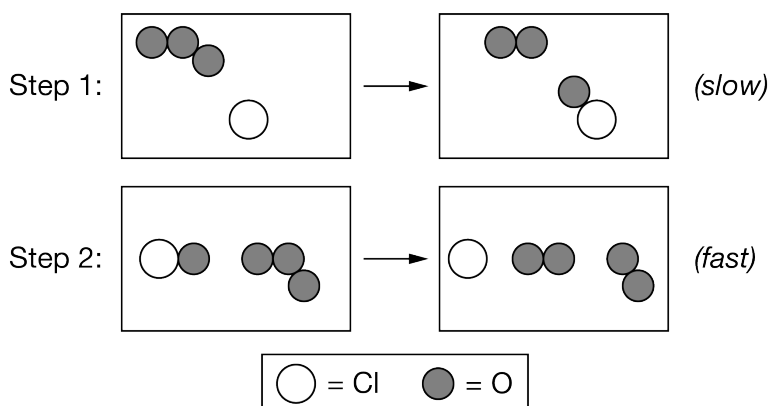


14. Which of the following rate laws for the overall reaction corresponds to the proposed mechanism?
- (A) $\text{Rate} = k[\text{O}_3]^2$
- (B) $\text{Rate} = k[\text{Cl}][\text{O}_3]$
- (C) $\text{Rate} = k[\text{ClO}][\text{O}_3]^2$
- (D) $\text{Rate} = k[\text{O}_2]^3 / [\text{O}_3]^2$
15. Which of the following reaction energy profiles best corresponds to the proposed mechanism?

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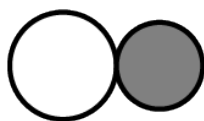


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The particle models shown above represent a proposed two-step mechanism for the destruction of ozone (O_3) in the upper atmosphere.

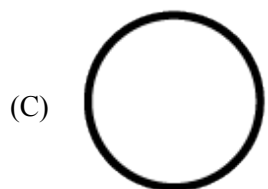
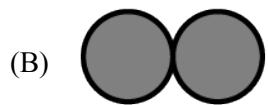
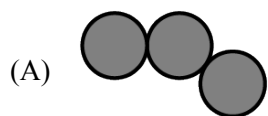
16.



Based on the proposed mechanism, which of the following best describes the concentration of the species represented above as the reaction occurs?

- (A) The concentration is larger than that of reactant or product.
- (B) The concentration is very low for the duration of the reaction.
- (C) The concentration increases steadily as the reaction proceeds.
- (D) The concentration fluctuates between high and low during the reaction.

17. Based on the models, which of the following represents a species that acts as a catalyst for the reaction?



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19. Based on the proposed mechanism, which of the following is the rate-law expression for the destruction of O_3 ?

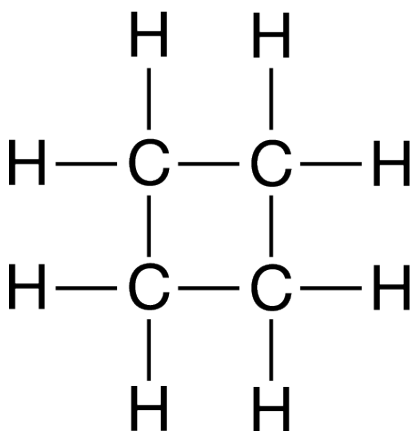
- (A) $\text{Rate} = k[\text{O}_3]^2$
 - (B) $\text{Rate} = k[\text{Cl}]^2$
 - (C) $\text{Rate} = k[\text{ClO}][\text{O}_3]$
 - (D) $\text{Rate} = k[\text{O}_3][\text{Cl}]$
-

18. Based on the proposed mechanism, what is the balanced chemical equation for the overall reaction?

- (A) $\text{O}_3(g) + \text{Cl}(g) \rightarrow \text{O}_2(g) + \text{ClO}(g)$
- (B) $2 \text{O}_3(g) + 2 \text{Cl}(g) \rightarrow 2 \text{O}_2(g) + 2 \text{ClO}(g)$
- (C) $\text{O}_3(g) + \text{Cl}(g) + \text{ClO}(g) \rightarrow 2 \text{O}_2(g) + \text{Cl}_2(g)$
- (D) $2 \text{O}_3(g) \rightarrow 3 \text{O}_2(g)$

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20. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



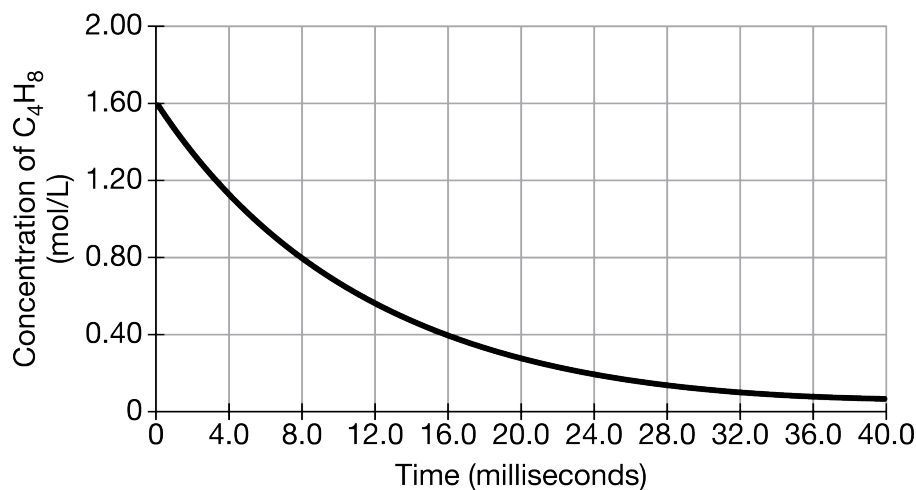
The hydrocarbon cyclobutane, C_4H_8 , is represented above. At high temperatures, cyclobutane quickly decomposes into ethene, C_2H_4 .

- (a) Draw a Lewis electron-dot diagram of the ethene molecule in the following box.



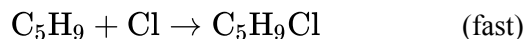
- (b) Estimate the value of the $\text{H} - \text{C} - \text{H}$ bond angle in ethene.
- (c) The graph below shows the concentration of cyclobutane as it decomposes over time at 1270 K.

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- (i) Based on the graph, determine the order of the decomposition reaction of cyclobutane at 1270 K. Justify your answer.
- (ii) Calculate the time, in milliseconds, that it would take for 99 percent of the original cyclobutane at 1270 K to decompose.

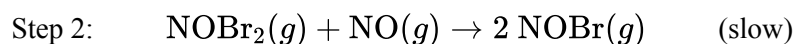
Another alkane with a ring structure is cyclopentane, $C_5H_{10}(g)$. Cyclopentane reacts with chlorine, $Cl_2(g)$, to produce $C_5H_9Cl(g)$ and $HCl(g)$. Following is a representation of a proposed mechanism for the reaction.



- (d) Write a rate law for the reaction that is consistent with the mechanism. Justify your answer.
- (e) A student claims that $Cl_2(g)$ is a catalyst in the reaction. Explain why the student's claim is false.



The reaction represented by the equation above has the following proposed mechanism.

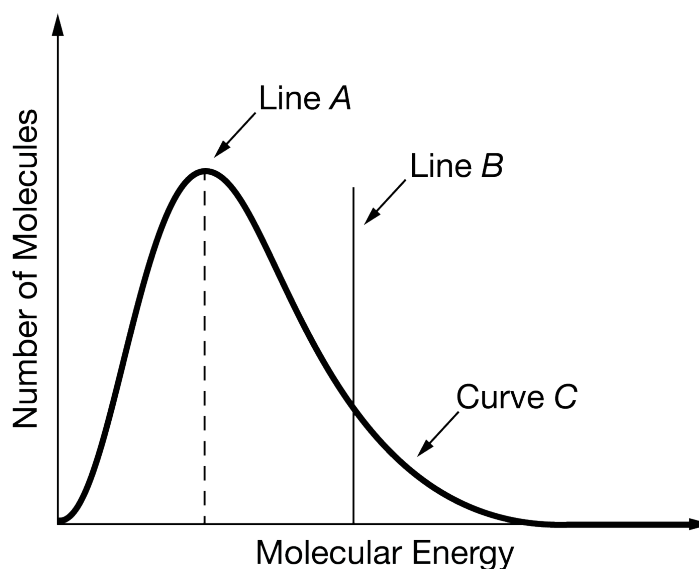


Based on the information, which of the following is the initial rate law for the reaction?

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- (A) $\text{Rate} = k[\text{Br}_2][\text{NO}]$
(B) $\text{Rate} = k[\text{Br}_2][\text{NO}]^2$
(C) $\text{Rate} = k[\text{Br}_2]^2[\text{NO}]$
(D) $\text{Rate} = k[\text{NOBr}_2][\text{NO}]$

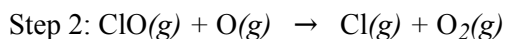
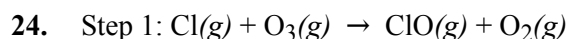
22.



The energy distribution profile (Curve C) for the Y_2 molecules is shown in the graph above for the reaction $\text{X} + \text{Y}_2 \rightarrow \text{XY}_2$ when it is done under certain experimental conditions. Line A represents the most probable energy of the Y_2 molecules, and Line B represents the activation energy. Which of the following changes should be made to the graph to explain the increase in the rate of the reaction if the only change in experimental conditions is the addition of a catalyst?

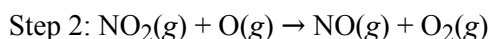
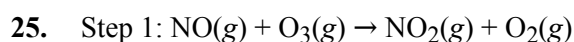
- (A) Curve C will broaden and Line B will move to the left, because more Y_2 molecules will have an energy greater than the minimum energy needed to overcome the activation energy barrier.
(B) Line B will move to the left because a larger fraction of the Y_2 molecules will have the minimum energy to overcome the activation energy barrier.
(C) Curve C and Line A will move to the right because the average energy of the Y_2 molecules will increase.
(D) Line A will move to the right because the most probable energy of the Y_2 molecules will increase.
23. To catalyze a biochemical reaction, an enzyme typically
- (A) drives the reaction to completion by consuming byproducts of the reaction
(B) binds temporarily to reactant molecules to lower the activation energy of the reaction
(C) dissociates into additional reactant molecules, thereby increasing the reaction rate
(D) decomposes and releases energy to increase the number of successful collisions between reactant molecules

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A proposed mechanism for destruction of ozone gas in the stratosphere is represented above. Which of the following is evidence that the mechanism is occurring?

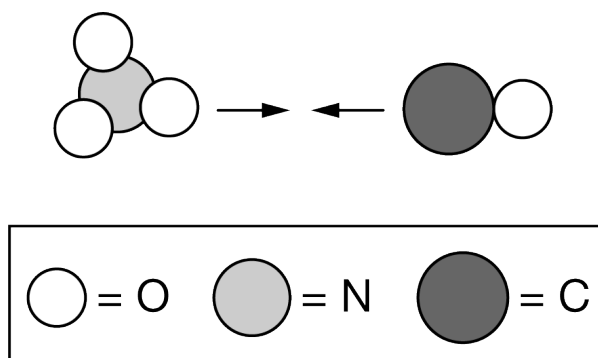
- (A) The presence of $\text{Cl}(g)$ increases the rate of the overall reaction.
- (B) The presence of $\text{Cl}(g)$ decreases the rate of the overall reaction.
- (C) The presence of $\text{Cl}(g)$ increases the equilibrium constant of the overall reaction.
- (D) The presence of $\text{Cl}(g)$ decreases the equilibrium constant of the overall reaction.



A reaction mechanism for the destruction of ozone, $\text{O}_3(g)$, is represented above. In the overall reaction, $\text{NO}(g)$ is best described as

- (A) an inhibitor
- (B) a catalyst
- (C) a reactant
- (D) an intermediate
- (E) a product

26.



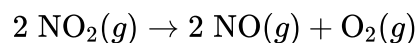
The slowest step in a reaction mechanism requires the collision represented above to occur. Which of the following most likely indicates how the addition of a solid catalyst could increase the rate of the reaction?

- (A) The catalyst could change the reaction from second order to third order.
- (B) The catalyst could increase the particles' speed, thereby increasing the value of the rate constant, k .
- (C) The catalyst could decrease the particles' speed, making it less likely that the particles will rebound without reacting when they collide.
- (D) The catalyst could adsorb one of the particles, making a successful (reaction-producing) collision with the other particle more likely.

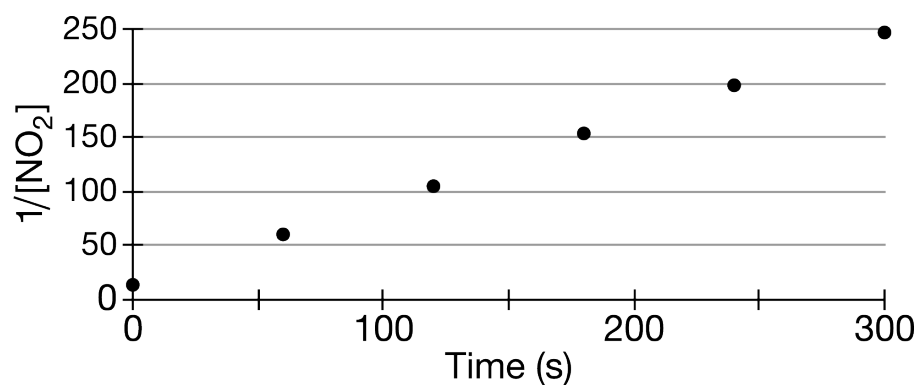
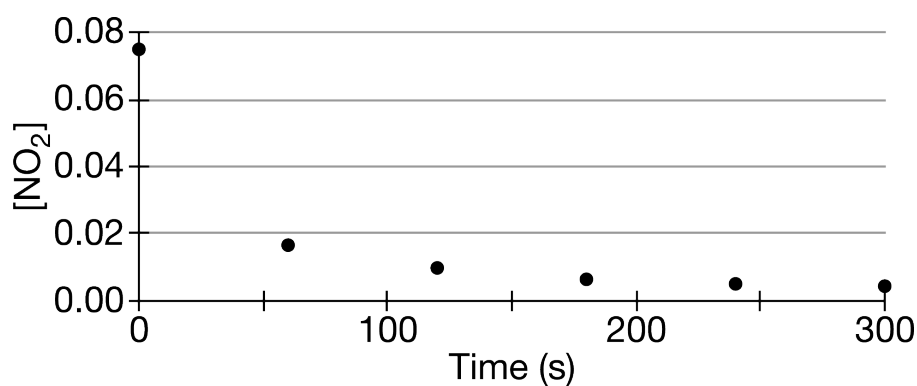
Chapter 09 Part 3: Reaction Mechanism

27. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

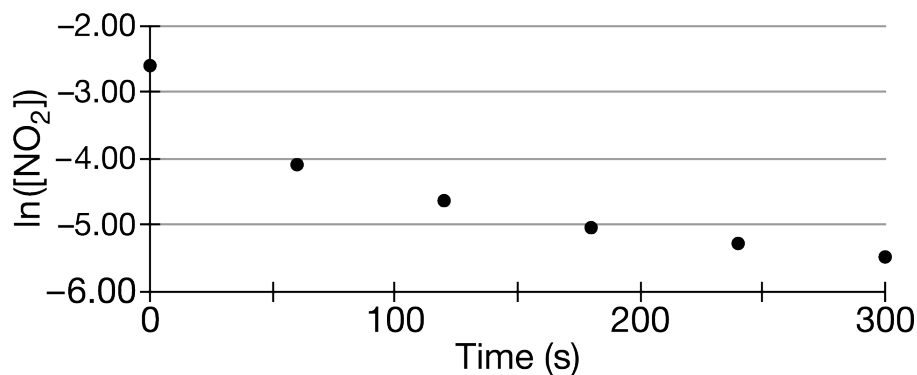
Nitrogen dioxide, $\text{NO}_2(g)$, is produced as a by-product of the combustion of fossil fuels in internal combustion engines. At elevated temperatures $\text{NO}_2(g)$ decomposes according to the equation below.



The concentration of a sample of $\text{NO}_2(g)$ is monitored as it decomposes and is recorded on the graph directly below. The two graphs that follow it are derived from the original data.

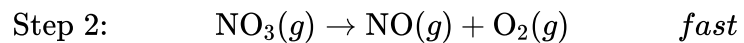
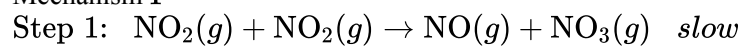


Chapter 09 Part 3: Reaction Mechanism



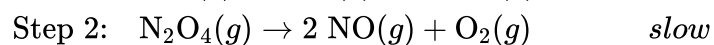
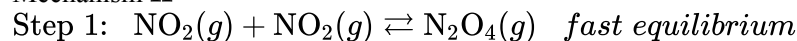
- (a) Explain how the graphs indicate that the reaction is second order.
- (b) Write the rate law for the decomposition of $\text{NO}_2(g)$.
- (c) Consider two possible mechanisms for the decomposition reaction.
- (i) Is the rate law described by mechanism I shown below consistent with the rate law you wrote in part (b) ? Justify your answer.

Mechanism I



- (ii) Is the rate law described by mechanism II shown below consistent with the rate law you wrote in part (b) ? Justify your answer.

Mechanism II



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28. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, L and X, are shown in the following table.

Compound	L	X
Diagram		
Boiling Point	371 K	437 K

- (a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.
- (b) The C — O bond in compound L is longer than any C — H bond in compound X. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound L —
Compound X —
- (d) Explain why the boiling point of compound X is higher than that of compound L based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A 0.10 mol sample of compound L is placed in an evacuated, rigid container at 400. K and 0.25 atm. When more moles of compound L are added, the pressure increases to 0.75 atm at 400. K. Determine the number of moles of compound L now present in the container.

Another carbon-containing compound is C₂H₆O, which reacts with H₂CrO₄, as represented by the following equation.



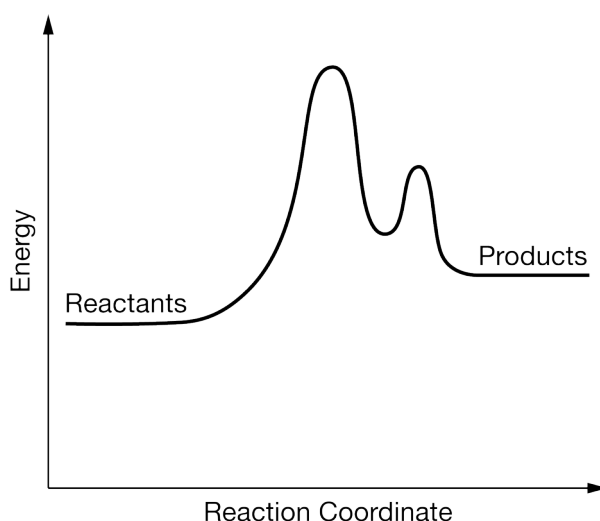
- (f) The student claims that C is oxidized when C₂H₆O is converted to C₂H₄O. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

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The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

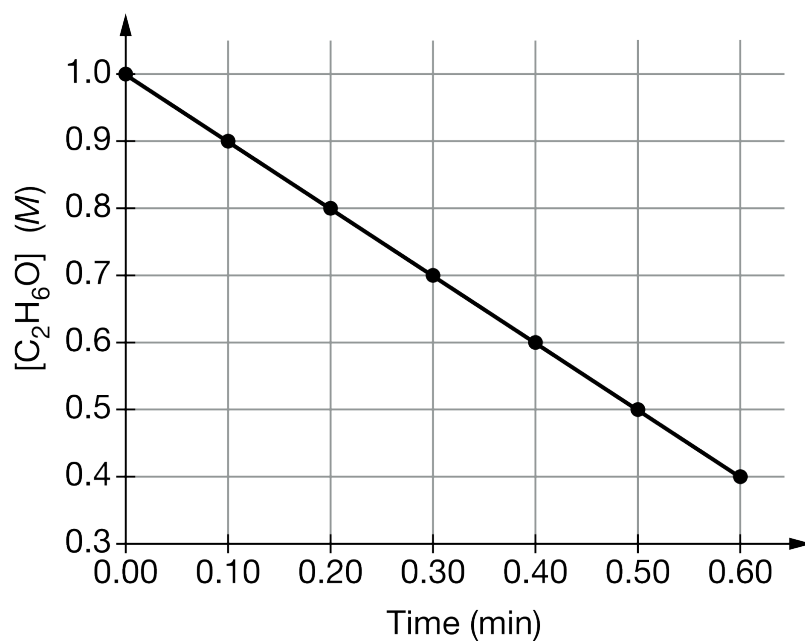
Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 1.0 M/min . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

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Chapter 09 Part 3: Reaction Mechanism

29. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, **L** and **Z**, are shown in the following table.

Compound	L	Z
Diagram	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \ddot{\text{O}} & - \text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & - \text{C} & - \text{C} & - \text{C} & - \text{C} & - \ddot{\text{S}} & - \text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $
Boiling Point	391 K	371 K

- (a) Based on VSEPR theory, what is the approximate value of the C — O — H bond angle in compound **L**?
- (b) The C — O bond in compound **L** is shorter than the C — S bond in compound **Z**. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound **L** —
Compound **Z** —
- (d) Explain why the boiling point of compound **L** is higher than that of compound **Z** based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound **Z** is placed in an evacuated, rigid container at a temperature of 400. K and a pressure of 0.50 atm. Determine the final pressure when the temperature of the sample is increased to 600. K.

Another carbon-containing compound is $\text{C}_2\text{H}_6\text{O}$, which reacts with H_2CrO_4 , as represented by the following equation.



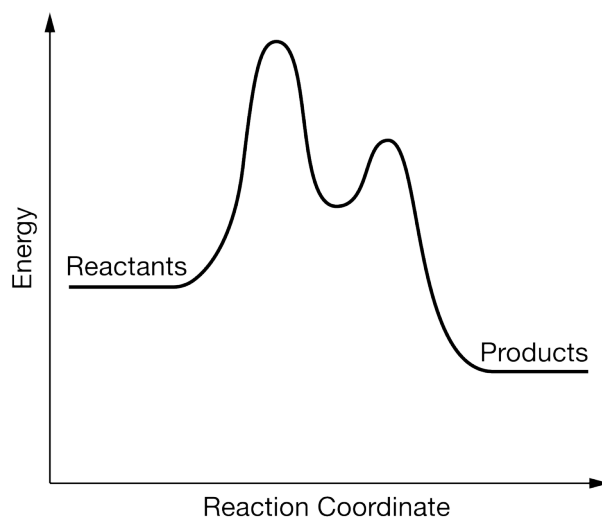
- (f) The student claims that O is oxidized when $\text{C}_2\text{H}_6\text{O}$ is converted to $\text{C}_2\text{H}_4\text{O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

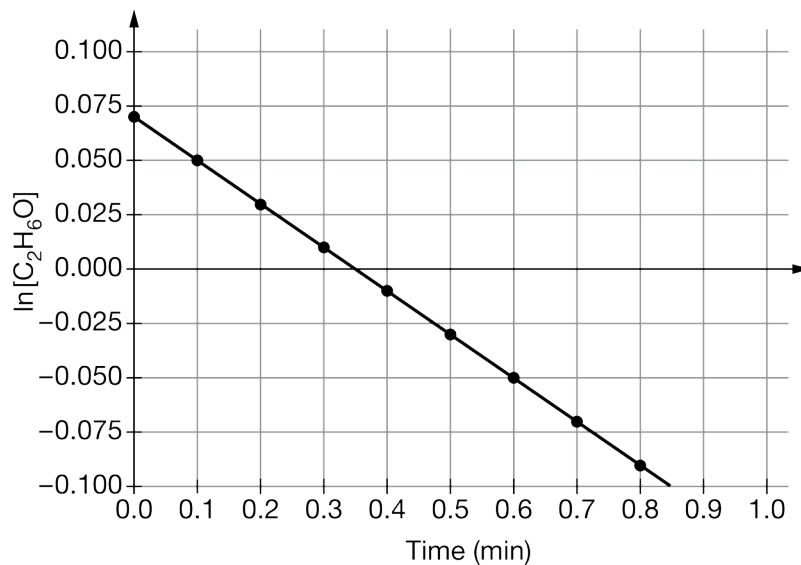
Chapter 09 Part 3: Reaction Mechanism

Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

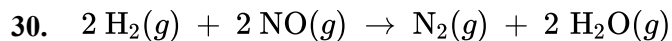
(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



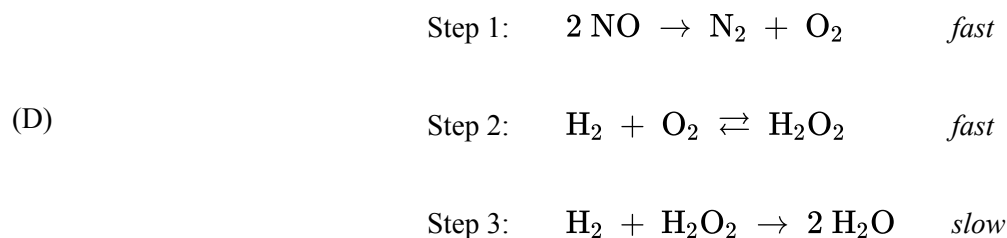
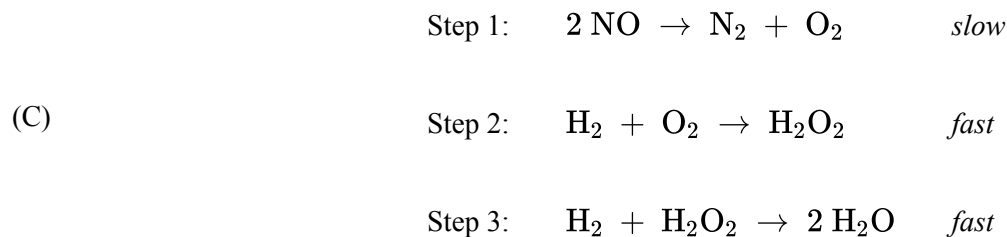
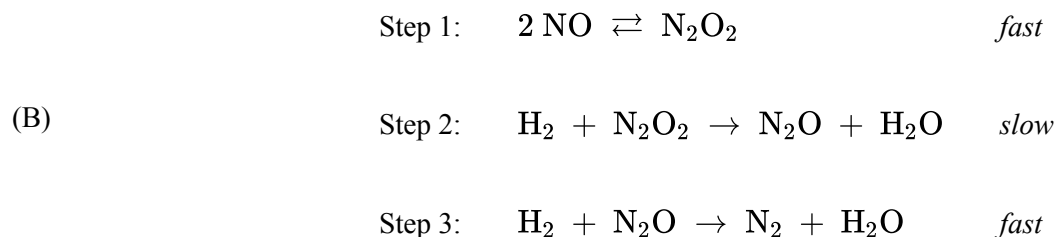
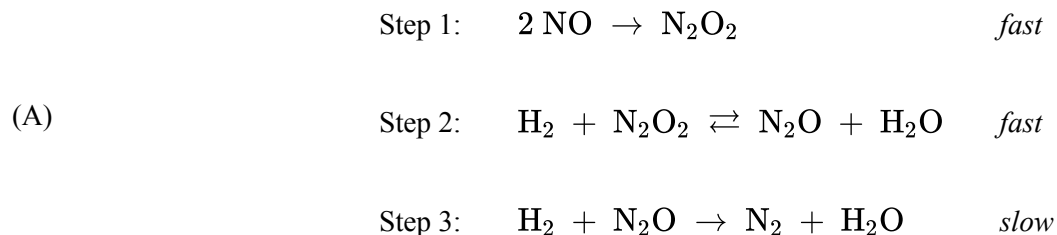
(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is first-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 0.50 min^{-1} . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



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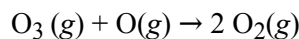


The experimental rate law for the reaction represented above is $\text{rate} = k[\text{H}_2][\text{NO}]^2$. Which of the following proposed mechanisms is consistent with the rate law?

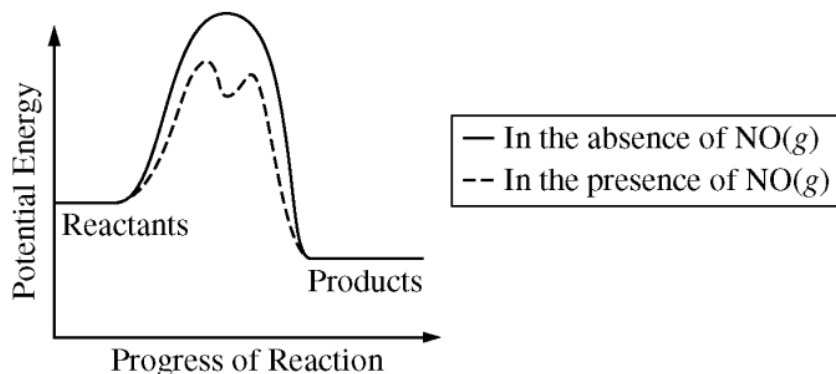


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31.



The decomposition of $\text{O}_3(g)$ in the upper atmosphere is represented by the equation above. The potential energy in the upper atmosphere is diagram for the decomposition of $\text{O}_3(g)$ in the presence and absence of $\text{NO}(g)$ is given below.



Which of the following mechanisms for the catalyzed reaction is consistent with the equation and diagram above?

- (A) $2 \text{O}_3(g) + 2 \text{NO}(g) \rightarrow 4 \text{O}_2(g) + \text{N}_2(g)$ *slow*
- (B) $\text{O}_3(g) + \text{NO}(g) \rightarrow \text{NO}_2(g) + \text{O}_2(g)$ *slow*
 $\text{NO}_2(g) + \text{O}(g) \rightarrow \text{NO}(g) + \text{O}_2(g)$ *fast*
- (C) $\text{NO}_2(g) + \text{O}_3(g) \rightarrow \text{NO}(g) + 2 \text{O}_2(g)$ *slow*
 $\text{NO}(g) + \text{O}(g) \rightarrow \text{NO}_2(g)$ *fast*
- (D) $\text{NO}_2(g) + \text{O}(g) \rightarrow \text{NO}_3(g)$ *slow*
 $\text{NO}_3(g) + \text{O}_3(g) \rightarrow \text{NO}_2(g) + 2 \text{O}_2(g)$ *fast*

32. $\text{N}_2\text{O}(g) + \text{CO}(g) \rightarrow \text{N}_2(g) + \text{CO}_2(g)$

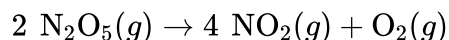
The rate of the reaction represented above increases significantly in the presence of $\text{Pd}(s)$. Which of the following best explains this observation?

- (A) Pd increases the activation energy of the reaction.
- (B) Pd absorbs the heat produced in the reaction.
- (C) One of the reactants binds on the surface of Pd , which introduces an alternative reaction pathway with a lower activation energy.
- (D) One of the products binds on the surface of Pd , which increases the reaction rate by decreasing the concentration of products in the mixture.

Chapter 09 Part 3: Reaction Mechanism

33. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

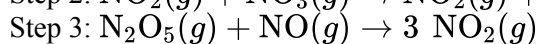
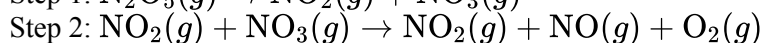
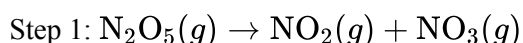
The following equation represents the decomposition of N_2O_5 , for which the rate law is $\text{rate} = k[\text{N}_2\text{O}_5]$.



A sample of pure $\text{N}_2\text{O}_5(g)$ is placed in an evacuated container and allowed to decompose at a constant temperature of 300 K. The concentration of $\text{N}_2\text{O}_5(g)$ in the container is measured over a period of time, and the measurements are recorded in the following table.

Time (hr)	$[\text{N}_2\text{O}_5](M)$
0	0.160
1.67	0.0800
3.33	0.0400
5.00	0.0200

- (a) Determine the value of the rate constant, k , for the reaction. Include units in your answer.
- (b) The following mechanism is proposed for the decomposition of $\text{N}_2\text{O}_5(g)$.



Identify which step of the proposed mechanism (1, 2, or 3) is the rate-determining step. Justify your answer in terms of the rate law given.

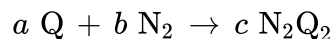
- (c) If this experiment was repeated at the same temperature but with twice the initial concentration of N_2O_5 , would the value of k increase, decrease, or remain the same? Explain your reasoning.

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34. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



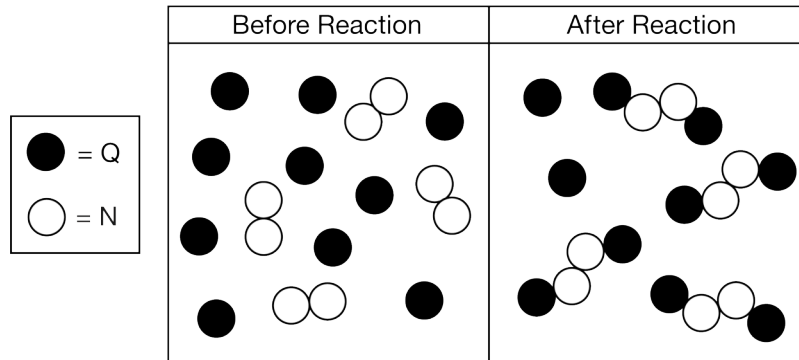
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

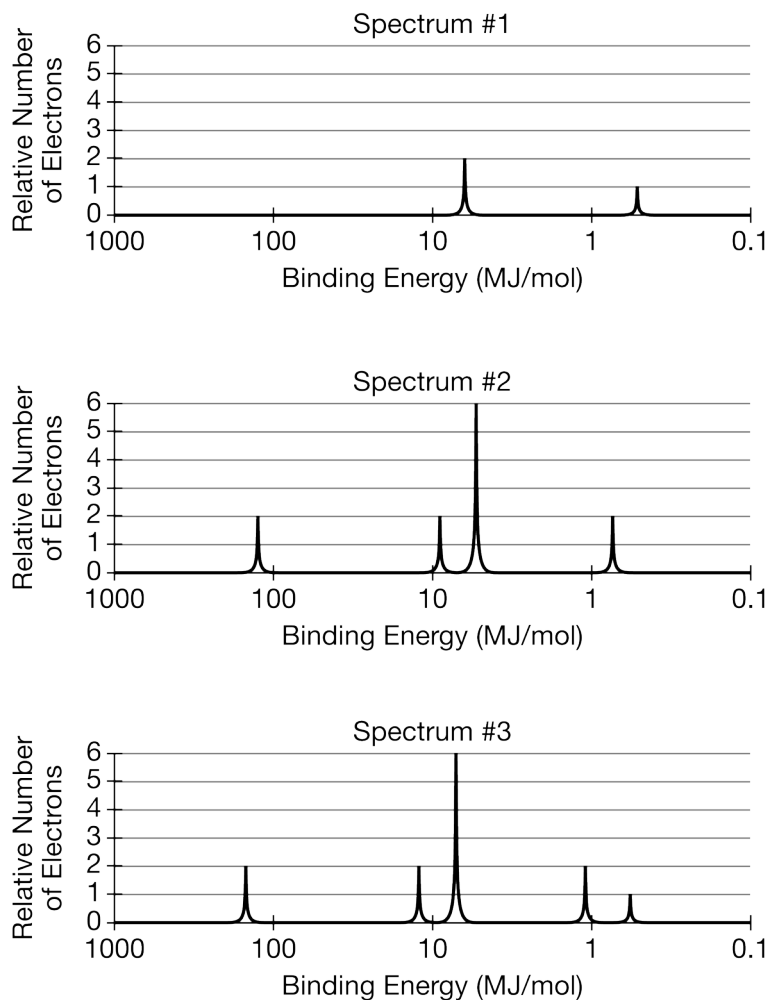


- (b) A student claims that N₂ is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound X₃N.

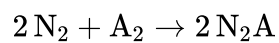
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

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(d) A student claims that the bonds in X_3N are nonpolar covalent. Do you agree or disagree with the student's hypothesis? Justify your choice.

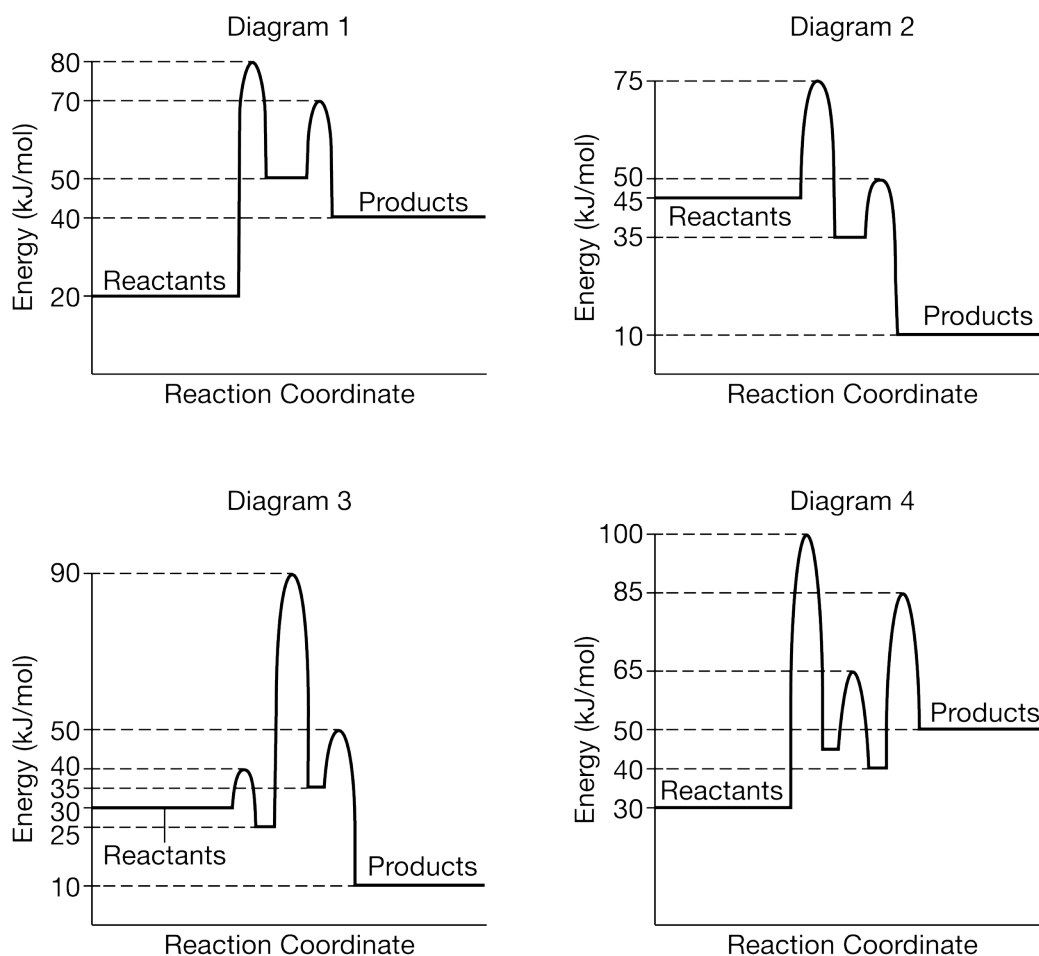
A new diatomic element, A_2 , is discovered and is found to have a molar mass of 52 g/mol. At high temperatures, elemental A_2 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A as shown in the equation below.



(e) Determine the mass of 0.50 mol of N_2A .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ < 0$.

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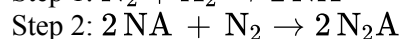
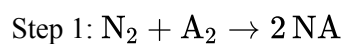


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound NA is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

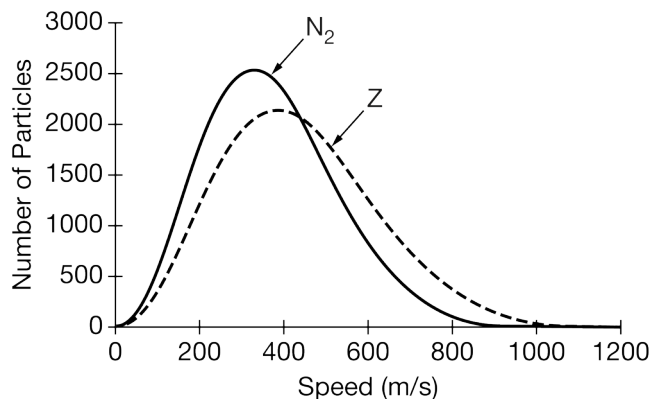
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N_2	77	5.6
Element R	89	6.8
Element Z	27	1.7

(j) Which sample, 2 moles of N_2 or 10 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, $\text{N}_2(l)$ or $\text{Z}(l)$, has the stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N_2 and element Z at the same temperature.



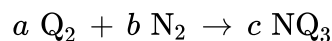
(l) Based on the data in the plot, is the molar mass of N_2 greater than, equal to, or less than the molar mass of Z? Justify your answer in terms of kinetic molecular theory.

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35. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure diatomic nonmetal, Q_2 , in a closed container containing pure nitrogen gas, N_2 . The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



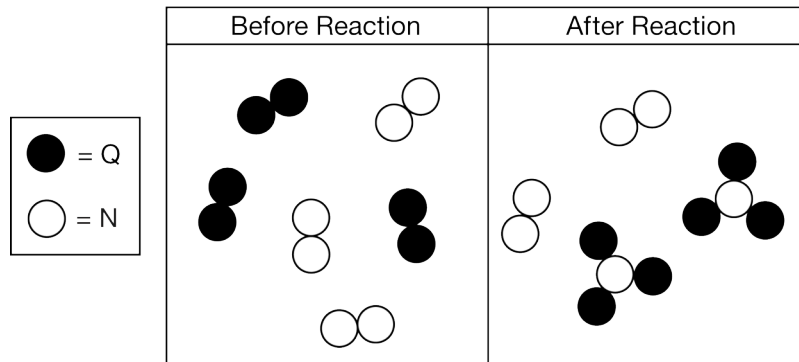
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

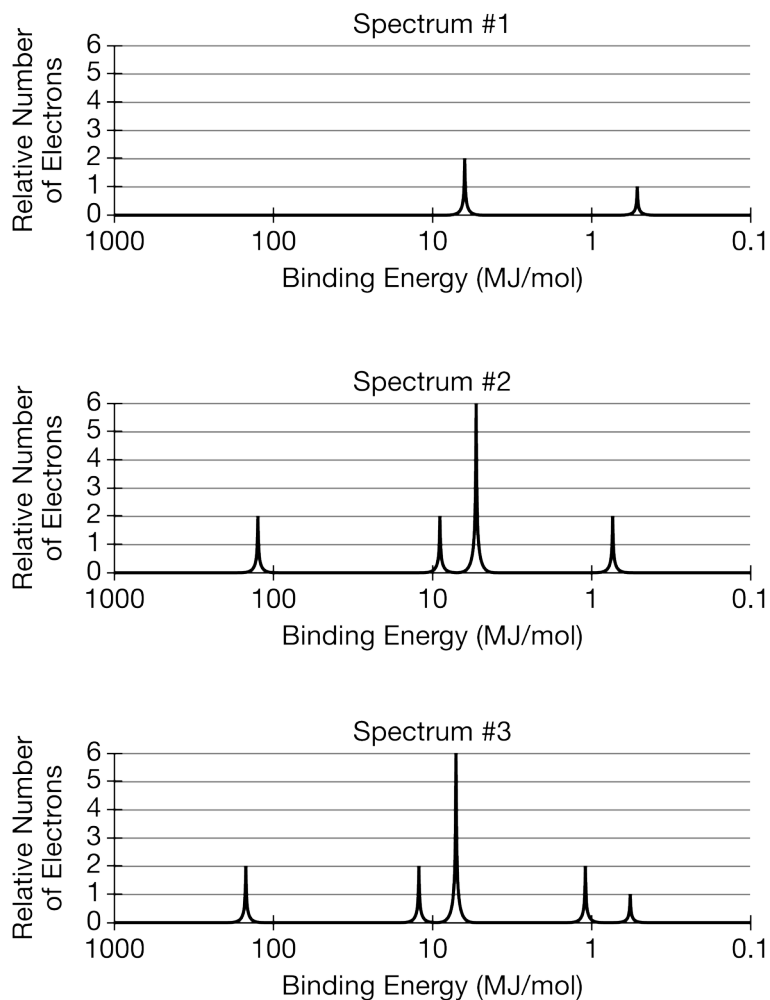


- (b) A student claims that Q_2 is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N_2 reacts with a pure metal X to form the compound X_3N_2 .

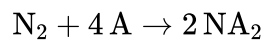
- (c) Which of the following complete photoelectron spectra could be that of metal X ? Justify your choice.

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(d) A student claims that the bonds in X_3N_2 are polar covalent. Do you agree or disagree with the student's claim? Justify your choice.

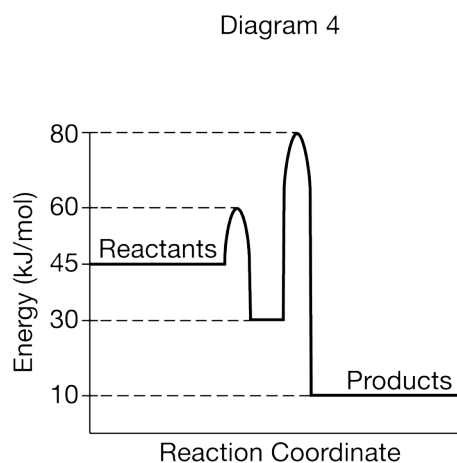
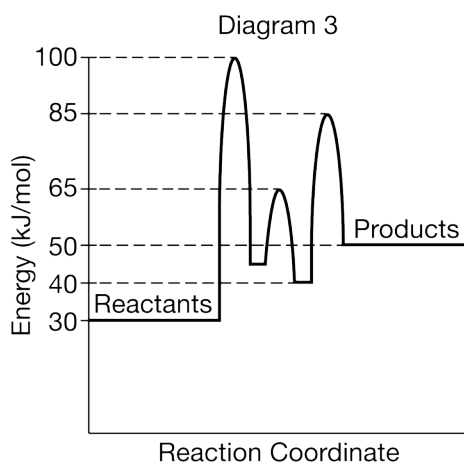
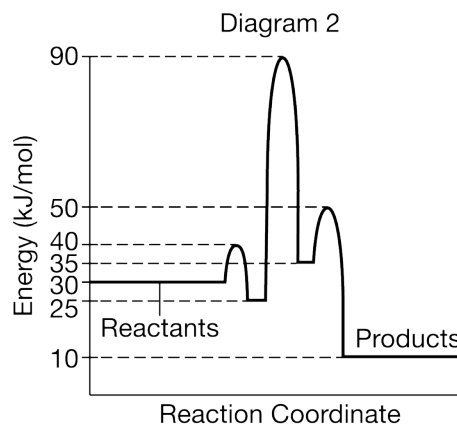
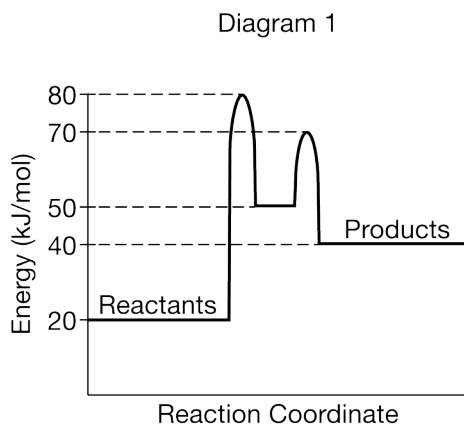
A new monatomic element, A, is discovered and is found to have a molar mass of 18 g/mol. At high temperatures, element A reacts with N_2 (molar mass 28 g/mol) to produce the compound NA_2 as shown in the equation below.



(e) Determine the mass of 0.25 mol of NA_2 .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ < 0$.

Chapter 09 Part 3: Reaction Mechanism

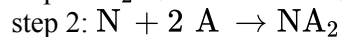


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the element L is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

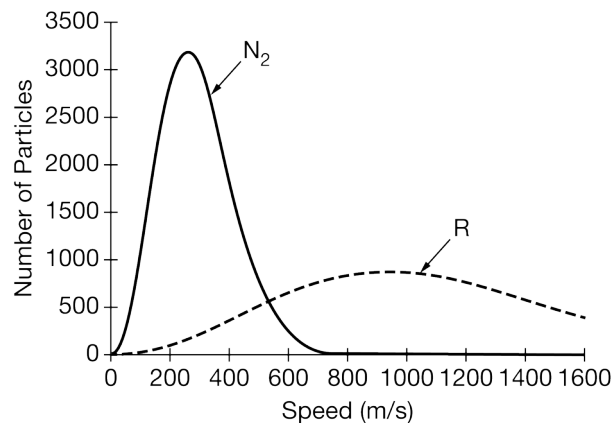
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point	Heat of Vaporization
	(K)	(kJ/mol)
Nitrogen, N_2	77	5.6
Element R	21	0.45
Element Z	165	12.6

(j) Which sample, 20 moles of R or 3 moles of N_2 , requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, $\text{N}_2(l)$ or $\text{R}(l)$, has stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N_2 and element R at the same temperature.



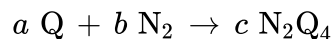
(l) Based on the data in the plot, is the molar mass of R greater than, equal to, or less than the molar mass of N_2 ? Justify your answer in terms of kinetic molecular theory.

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36. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N_2 . The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



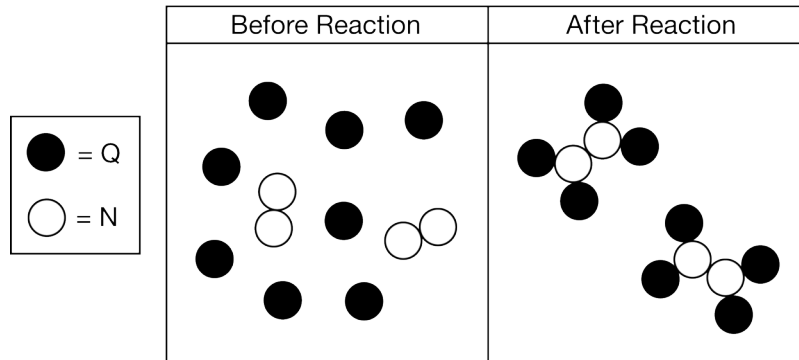
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

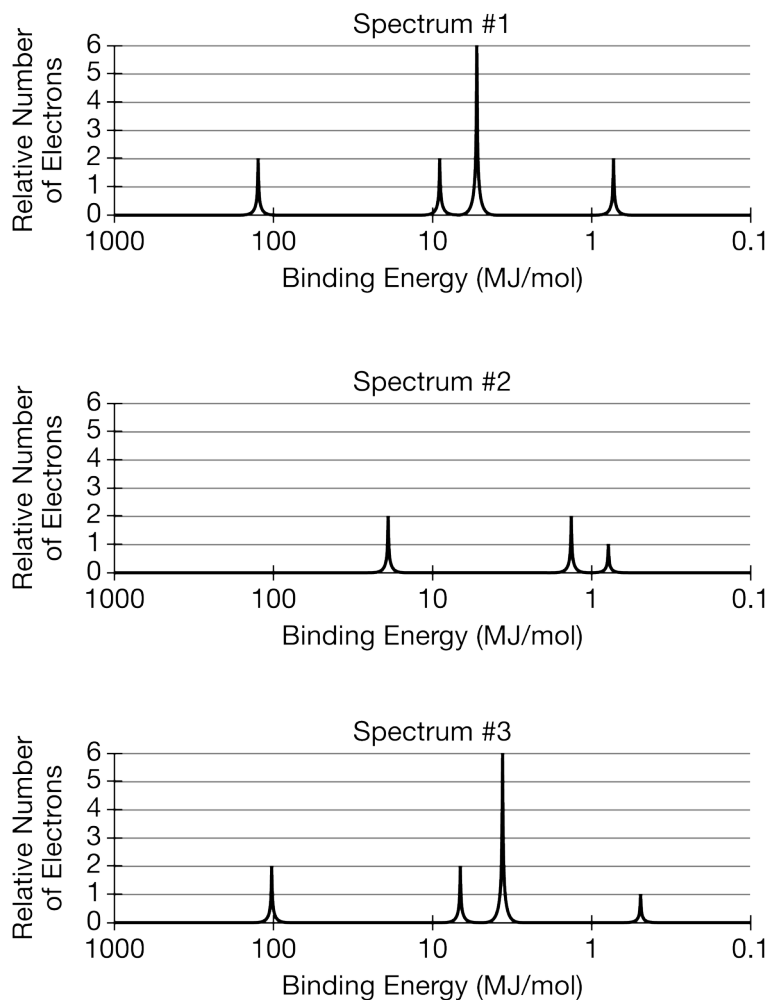


- (b) A student claims that N_2 is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N_2 reacts with a pure metal X to form the compound X_3N_2 .

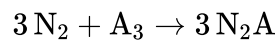
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

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(d) A student claims that molecule X_3N_2 contains both ionic and polar covalent bonds. Do you agree or disagree with the student's claim? Justify your choice.

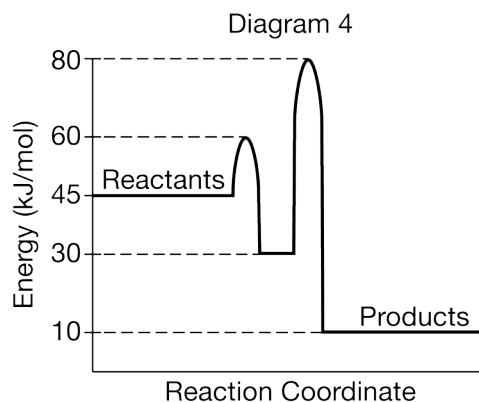
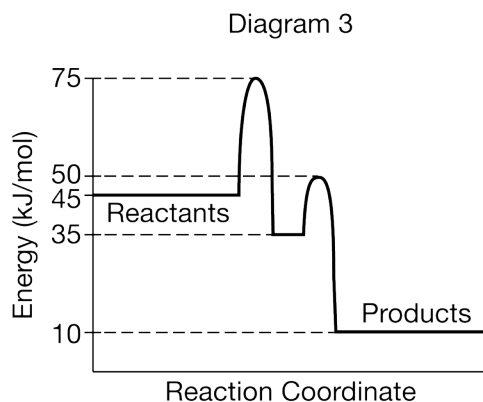
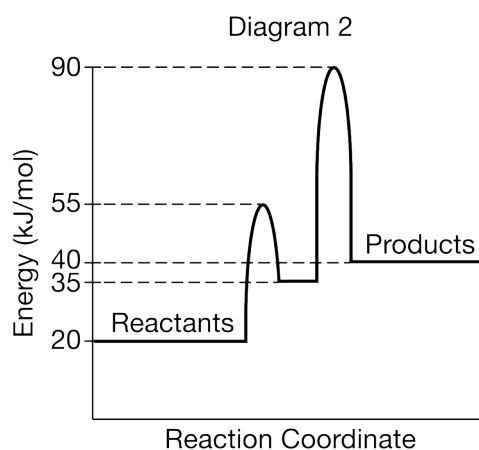
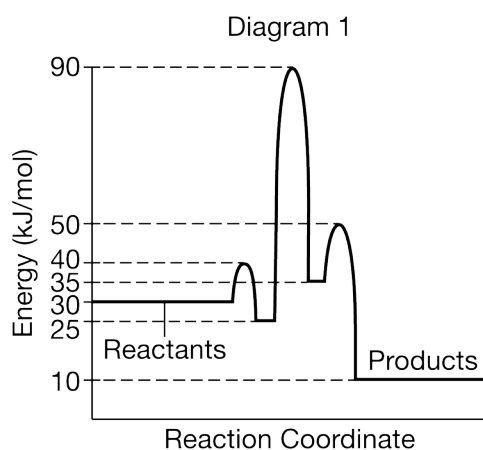
A new triatomic element, A_3 , is discovered and is found to have a molar mass of 93 g/mol. At high temperatures, A_3 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A as shown in the equation below.



(e) Determine the mass of 0.25 mol of N_2A .

The reaction above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ > 0$.

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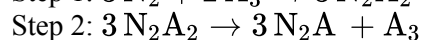
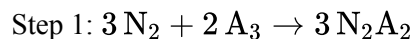


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that element A_3 is an intermediate. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

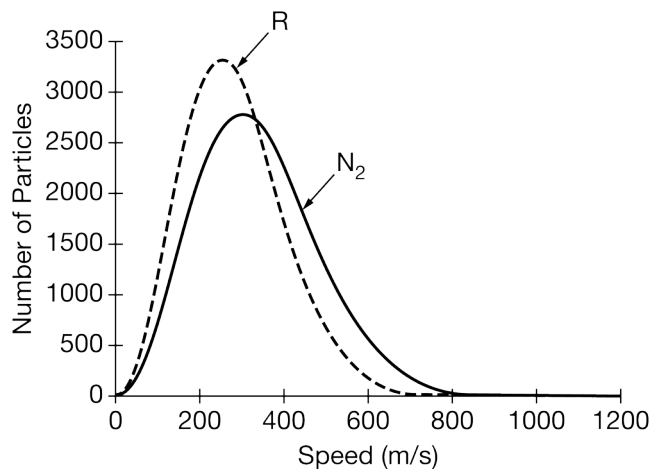
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N_2	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 3 moles of R or 1 mole of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, R(l) or $\text{N}_2(l)$, has weaker attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N_2 and element R at the same temperature.



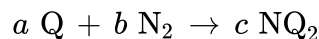
(l) Based on the data in the plot, is the molar mass of R greater than, equal to, or less than the molar mass of N_2 ? Justify your answer in terms of kinetic molecular theory.

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37. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



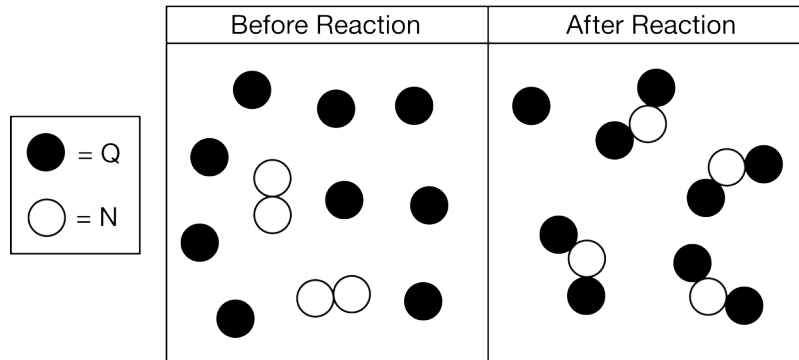
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

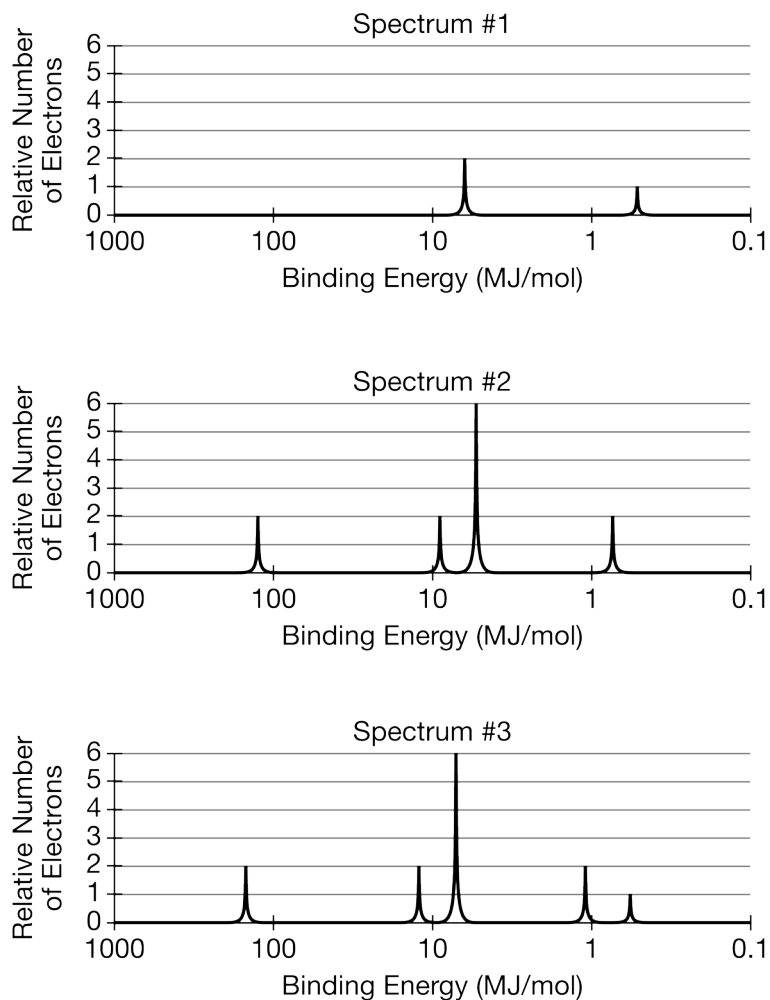


- (b) A student claims that N₂ is the excess reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound XN.

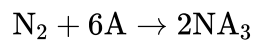
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

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(d) A student claims that the bonds in XN are ionic. Do you agree or disagree with the student's claim? Justify your choice.

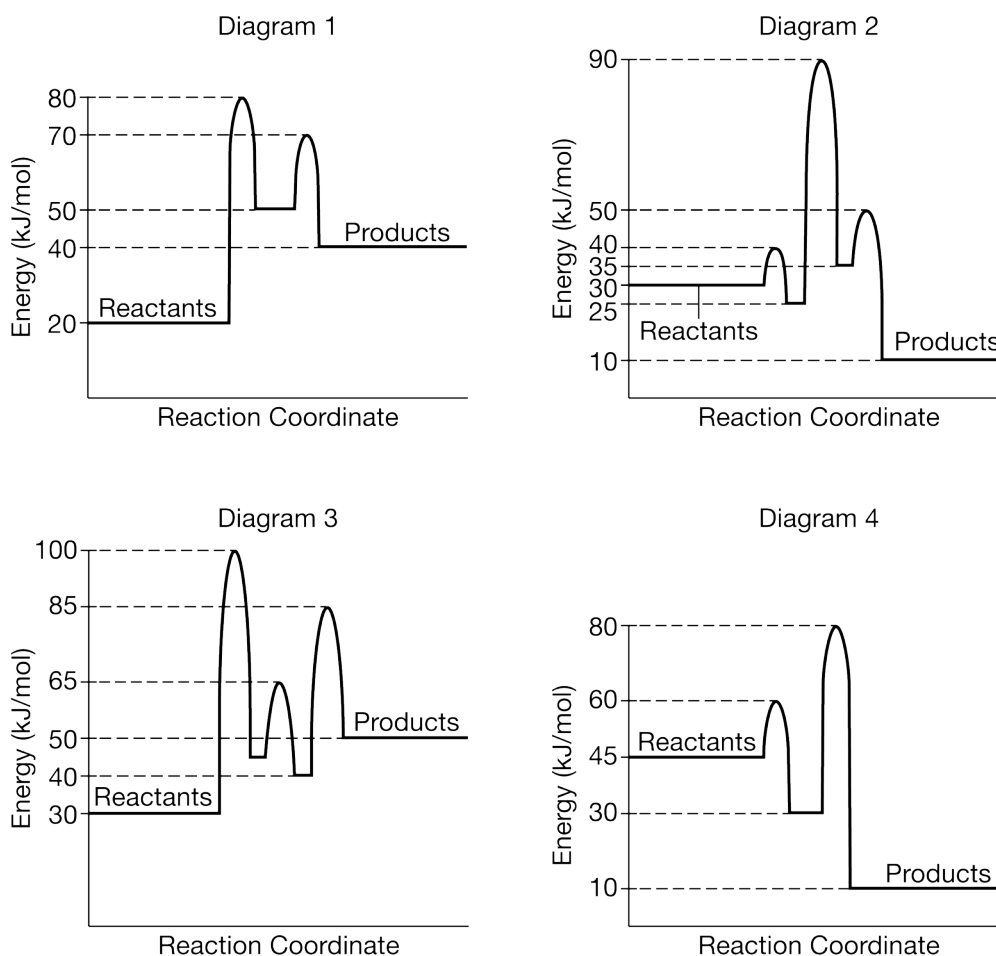
A new monatomic element, A , is discovered and is found to have a molar mass of 13 g/mol . At high temperatures, element A reacts with N_2 (molar mass 28 g/mol) to produce the compound NA_3 as shown in the equation below.



(e) Determine the mass of 0.50 mol of NA_3 .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{\text{rxn}}^\circ > 0$.

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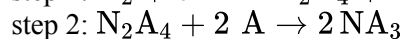
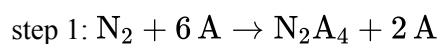


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound N_2A_4 is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

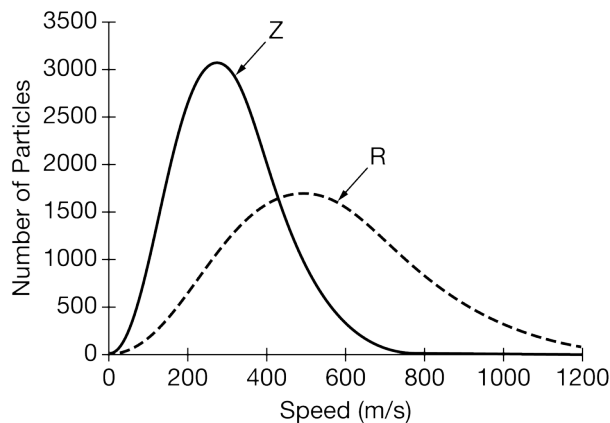
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N_2	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 5 moles of R or 3 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, $\text{Z}(l)$ or $\text{R}(l)$, has stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of elements Z and R at the same temperature.



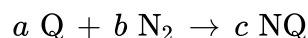
(l) Based on the data in the plot, is the molar mass of Z greater than, equal to, or less than the molar mass of R? Justify your answer in terms of kinetic molecular theory.

Chapter 09 Part 3: Reaction Mechanism

38. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



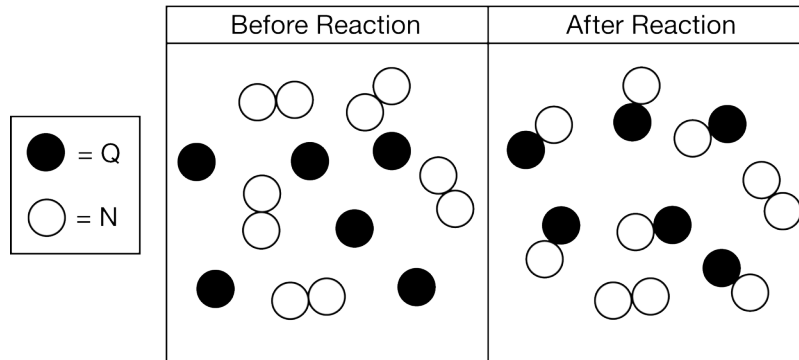
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

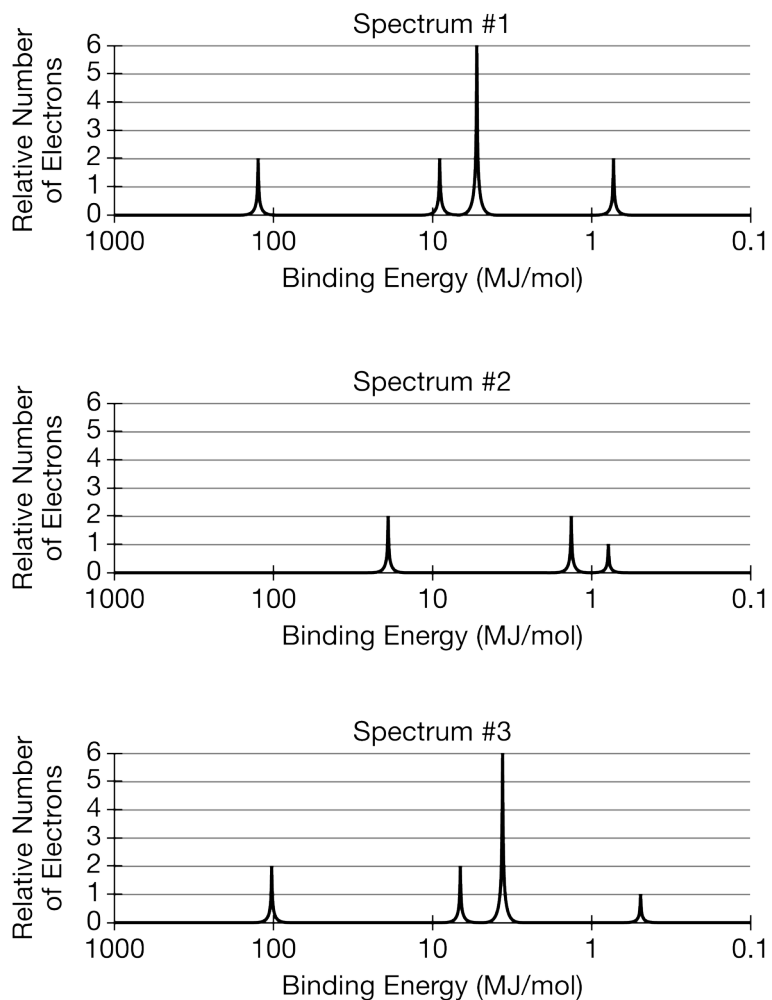


- (b) A student claims that Q is the excess reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound X₃N.

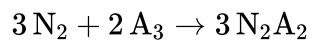
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

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(d) A student claims that molecule X_3N contains both ionic and nonpolar covalent bonds. Do you agree or disagree with the student's claim? Justify your choice.

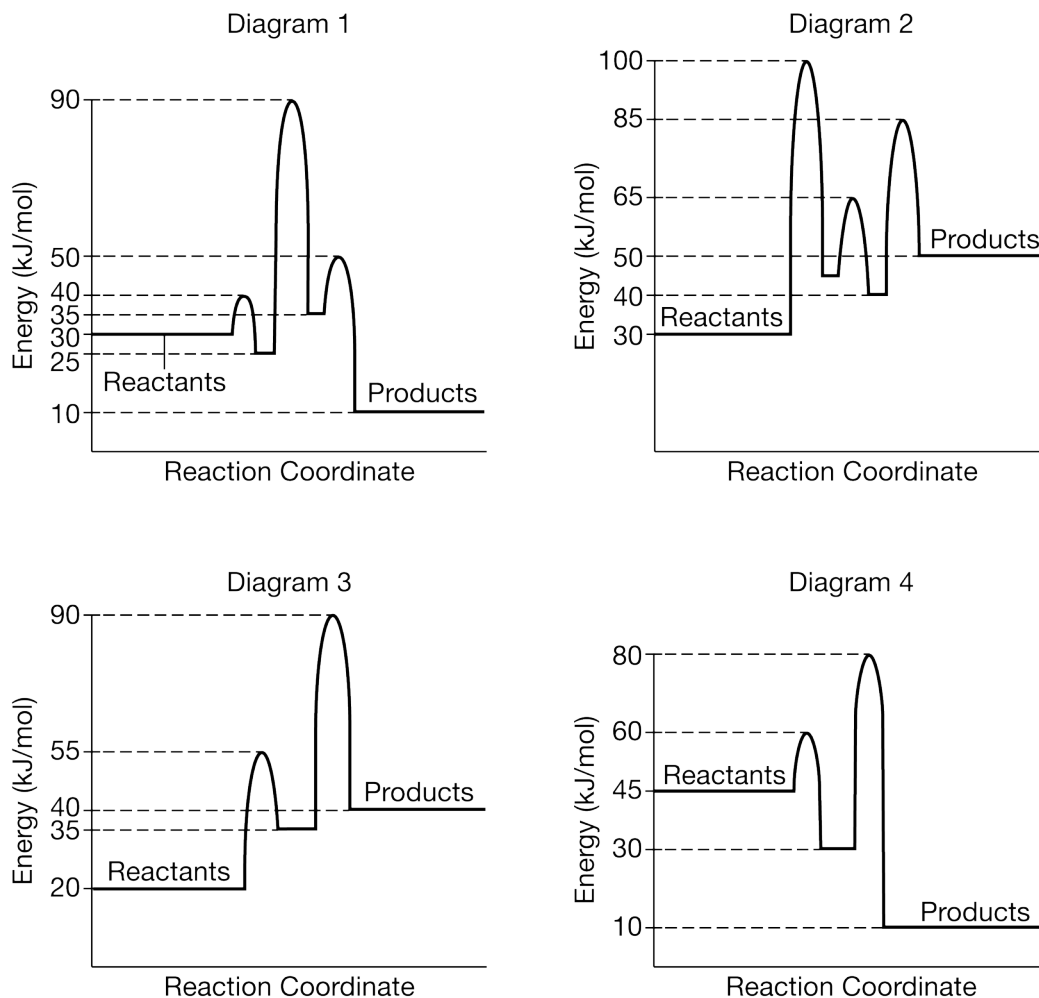
A new triatomic element, A_3 , is discovered and is found to have a molar mass of 45 g/mol. At high temperatures, A_3 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A_2 as shown in the equation below.



(e) Determine the mass of 0.50 mol of N_2A_2 .

The reaction above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ > 0$.

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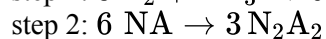
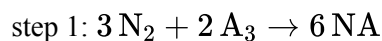


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound NA is an intermediate. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

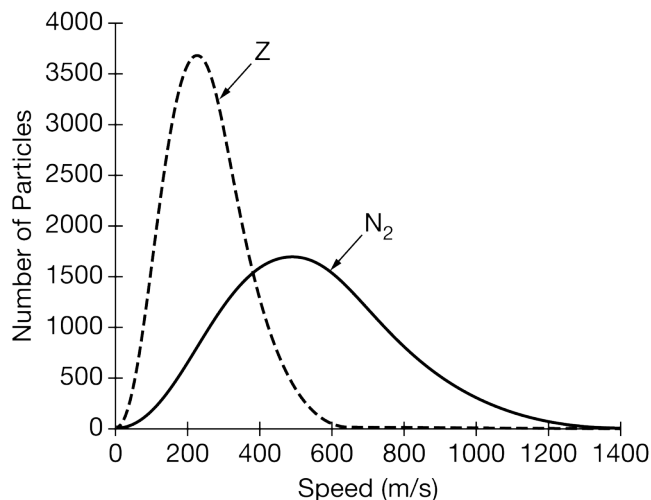
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N_2	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 5 moles of N_2 or 2 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, $\text{Z}(l)$ or $\text{N}_2(l)$, has weaker attractions among its particles? Justify your answer using information from the table.

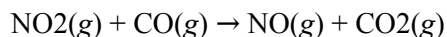
The plot below shows the distribution of particle speeds for equimolar amounts of N_2 and element Z at the same temperature.



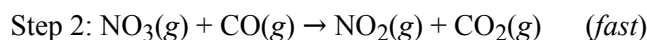
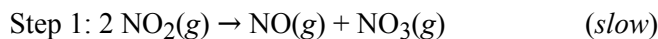
(l) Based on the data in the plot, is the molar mass of Z greater than, equal to, or less than the molar mass of N_2 ? Justify your answer in terms of kinetic molecular theory.

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39.



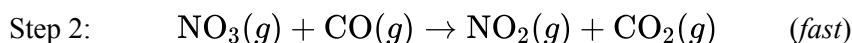
The reaction between $\text{NO}_2(g)$ and $\text{CO}(g)$ is represented above. The elementary steps of a proposed reaction mechanism are represented below.



Which of the following is the rate law for the overall reaction that is consistent with the proposed mechanism?

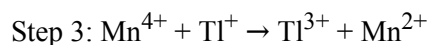
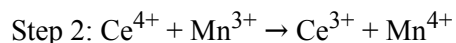
- (A) $\text{Rate} = k [\text{NO}_2][\text{CO}]$
- (B) $\text{Rate} = k [\text{NO}_2]^2$
- (C) $\text{Rate} = k [\text{NO}_3][\text{CO}]$
- (D) $\text{Rate} = k [\text{NO}_2][\text{NO}_3][\text{CO}]$

40.



A two-step reaction mechanism is proposed for a gas-phase reaction, as represented above. Which of the following correctly identifies both the chemical equation for step 1 and the rate law for the overall reaction?

- (A) The chemical equation for step 1 is $2 \text{NO}_2(g) \rightarrow \text{NO}(g) + \text{NO}_3(g)$, and the rate law is $\text{rate} = [\text{NO}_2]^2$.
- (B) The chemical equation for step 1 is $\text{NO}_2(g) + \text{CO}(g) \rightarrow \text{NO}(g) + \text{CO}_2(g)$, and the rate law is $\text{rate} = [\text{NO}_2][\text{CO}]$.
- (C) The chemical equation for step 1 is $\text{NO}_3(g) + 2 \text{CO}(g) \rightarrow \text{NO}(g) + 2 \text{CO}_2(g)$, and the rate law is $\text{rate} = [\text{NO}_2][\text{CO}]$.
- (D) The chemical equation for step 1 is $\text{NO}_3(g) + \text{NO}_2(g) + 2 \text{CO}(g) \rightarrow \text{NO}(g) + \text{NO}_2(g) + 2 \text{CO}_2(g)$, and the rate law is $\text{rate} = [\text{NO}_3][\text{NO}_2][\text{CO}]^2$.



The proposed steps for a catalyzed reaction between Ce^{4+} and Ti^+ are represented above. The products of the overall catalyzed reaction are

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- (A) Ce^{4+} and Tl^{+}
- (B) Ce^{3+} and Tl^{3+}
- (C) Ce^{3+} and Mn^{3+}
- (D) Ce^{3+} and Mn^{4+}
- (E) Tl^{3+} and Mn^{2+}

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42. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, Q and X, are shown in the following table.

Compound	Q	X
Diagram	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad :\text{O}: \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \quad \quad \uparrow \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
Boiling Point	281 K	437 K

- (a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.
- (b) The C — H bonds in compound Q are shorter than the C — C bonds in compound X. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound Q –
Compound X –
- (d) Explain why the boiling point of compound X is higher than that of compound Q based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound Q is placed in an evacuated cylinder with a movable piston that can change the volume of the cylinder. The initial volume and temperature of the gas are 10.0 L and 400. K. If the temperature of the gas is raised to 500. K as its pressure remains constant, what is the final volume of the gas?

Another carbon-containing compound is C₂H₆O, which reacts with H₂CrO₄, as represented by the following equation.



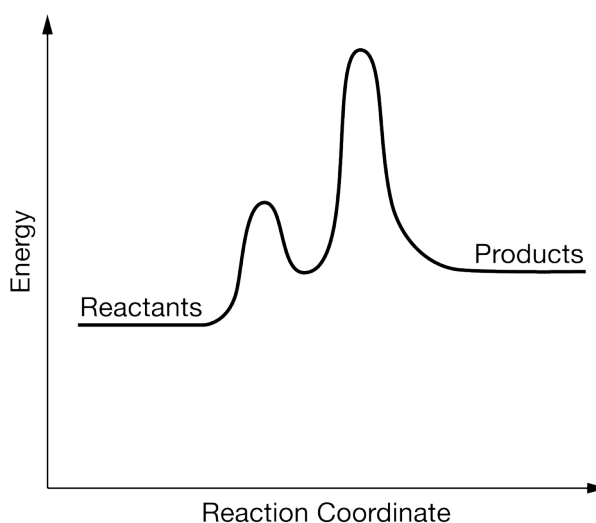
- (f) The student claims that Cr is reduced when H₂CrO₄ is converted to H₂CrO₃. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

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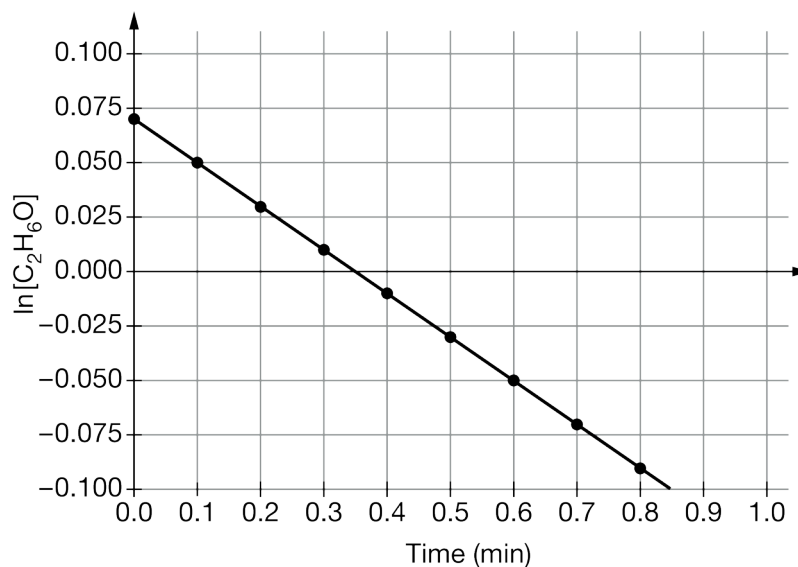
The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is second-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is $0.20 \text{ M}^{-1} \text{ min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



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43. $\text{NO}(g)$ and $\text{O}_2(g)$ react to form $\text{NO}_2(g)$. The rate law of the reaction is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$. If the reaction occurs in a single elementary step that is a three-body molecular collision, then which of the following is the equation for the elementary step?

- (A) $\text{N} + \text{O} + \text{O} \rightarrow \text{NO}_2$
(B) $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$
(C) $\text{N} + \text{O}_2 \rightarrow \text{NO}_2$
(D) $\text{N}_2 + 2\text{O}_2 \rightarrow 2\text{NO}_2$

- 44.
- | | | |
|---------|--|--------|
| Step 1: | $\text{NO}_2(g) + \text{F}_2(g) \rightleftharpoons \text{NO}_2\text{F}_2(g)$ | (fast) |
| Step 2: | $\text{NO}_2\text{F}_2(g) \rightarrow \text{NO}_2\text{F}(g) + \text{F}(g)$ | (slow) |
| Step 3: | $\text{F}(g) + \text{NO}_2(g) \rightarrow \text{NO}_2\text{F}(g)$ | (fast) |

A proposed mechanism for the chemical reaction $2\text{NO}_2(g) + \text{F}_2(g) \rightarrow 2\text{NO}_2\text{F}(g)$ is shown above. Which of the following rate laws is consistent with this mechanism?

- (A) $\text{Rate} = k[\text{NO}_2][\text{F}_2]$
(B) $\text{Rate} = k[\text{NO}_2]^2[\text{F}_2]$
(C) $\text{Rate} = k[\text{NO}_2][\text{F}_2]^2$
(D) $\text{Rate} = k[\text{NO}_2]^2[\text{F}_2]^2$
45. $2\text{HBr}(g) + \text{O}_2(g) \rightarrow \text{H}_2\text{O}_2(g) + \text{Br}_2(g)$

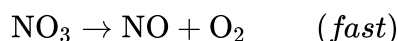
Based on a kinetics study of the reaction represented by the equation above, the following mechanism for the reaction is proposed.

- | | | |
|---------|--|--------|
| Step 1: | $\text{HBr}(g) + \text{O}_2(g) \rightarrow \text{HO}_2\text{Br}(g)$ | (slow) |
| Step 2: | $\text{HO}_2\text{Br}(g) + \text{HBr}(g) \rightarrow 2\text{HOBr}(g)$ | (fast) |
| Step 3: | $2\text{HOBr}(g) \rightarrow \text{H}_2\text{O}_2(g) + \text{Br}_2(g)$ | (fast) |

Which of following rate laws is consistent with the proposed mechanism?

- (A) $\text{Rate} = k[\text{HBr}]^2[\text{O}_2]$
(B) $\text{Rate} = k[\text{HBr}][\text{O}_2]$
(C) $\text{Rate} = k[\text{HO}_2\text{Br}][\text{HBr}]$
(D) $\text{Rate} = k[\text{HOBr}]^2$

Chapter 09 Part 3: Reaction Mechanism



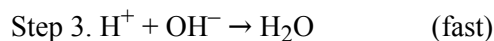
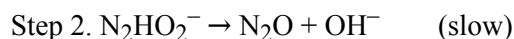
A proposed two-step mechanism for the reaction $2 \text{NO}_2 \rightarrow 2 \text{NO} + \text{O}_2$ is represented. Which of the following rate laws is consistent with the mechanism?

(A) $\text{rate} = k[\text{NO}_3]$

(B) $\text{rate} = k[\text{NO}_2]^2$

(C) $\text{rate} = k[\text{NO}_2]^2[\text{NO}_3]$

(D) $\text{rate} = k \frac{[\text{NO}]^2[\text{O}_2]}{[\text{NO}_2]^2}$



Nitramide, $\text{N}_2\text{H}_2\text{O}_2$, decomposes slowly in aqueous solution. This decomposition is believed to occur according to the reaction mechanism above. The rate law for the decomposition of nitramide that is consistent with this mechanism is given by which of the following?

(A) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2]$

(B) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2] [\text{H}^+]$

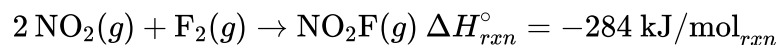
(C) $\text{Rate} = k \frac{[\text{N}_2\text{H}_2\text{O}_2]}{[\text{H}^+]}$

(D) $\text{Rate} = k \frac{[\text{N}_2\text{H}_2\text{O}_2]}{[\text{N}_2\text{HO}_2^-]}$

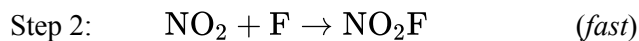
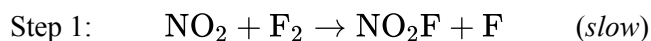
(E) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2] [\text{OH}^-]$

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48. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



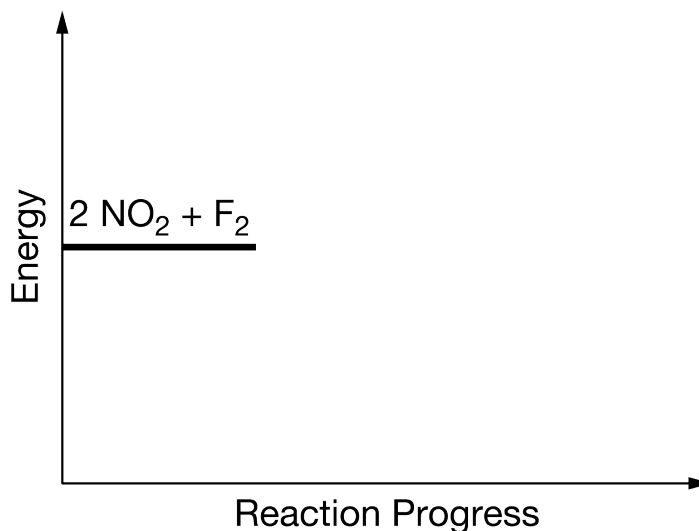
$\text{NO}_2(g)$ and $\text{F}_2(g)$ can react to produce $\text{NO}_2\text{F}(g)$, as represented above. A proposed mechanism for the reaction has two elementary steps, as shown below.



- (a) Write a rate law for the overall reaction that is consistent with the proposed mechanism.
- (b) On the incomplete reaction energy diagram below, draw a curve that shows the following two details.

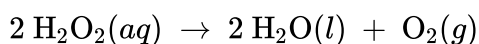
The relative activation energies of the two elementary steps

The enthalpy change of the overall reaction



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The following balanced equation represents the decomposition of hydrogen peroxide, H_2O_2 .



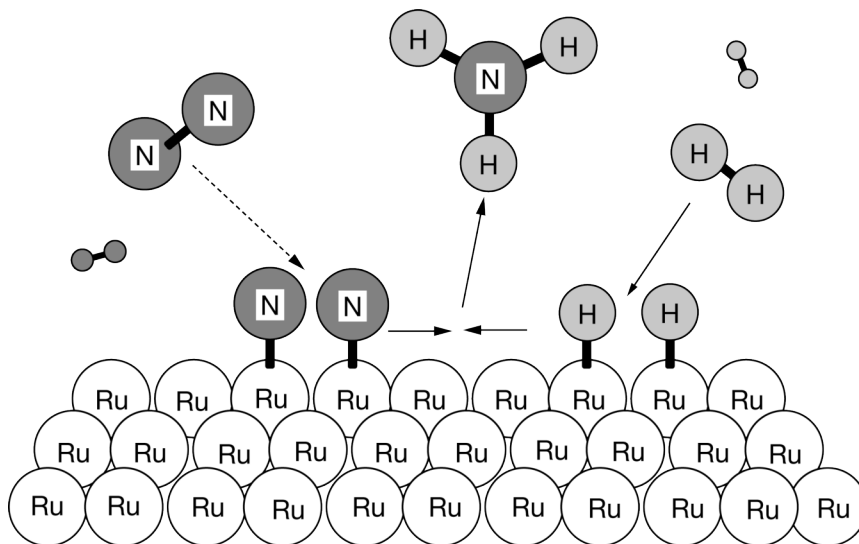
A student carries out an experiment in which a sample of $\text{H}_2\text{O}_2(aq)$ decomposes in the presence of $\text{MnO}_2(s)$. The student measures the volume of $\text{O}_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

Time (s)	Volume of $\text{O}_2(g)$ Collected (mL)
0	0
60	42
120	64
180	76
240	82
300	86
360	88
420	89

49. The student observes that under the same conditions, the reaction occurs at a much slower rate when $\text{MnO}_2(s)$ is not used. Which of the following statements best explains this observation?
- (A) $\text{MnO}_2(s)$ changes the reaction from endothermic to exothermic.
- (B) $\text{MnO}_2(s)$ shifts the equilibrium position of the reaction from reactant favored to product favored.
- (C) $\text{MnO}_2(s)$ releases heat in order to start the reaction.
- (D) $\text{MnO}_2(s)$ provides a reaction pathway with a lower activation energy.
-

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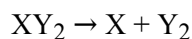
50.



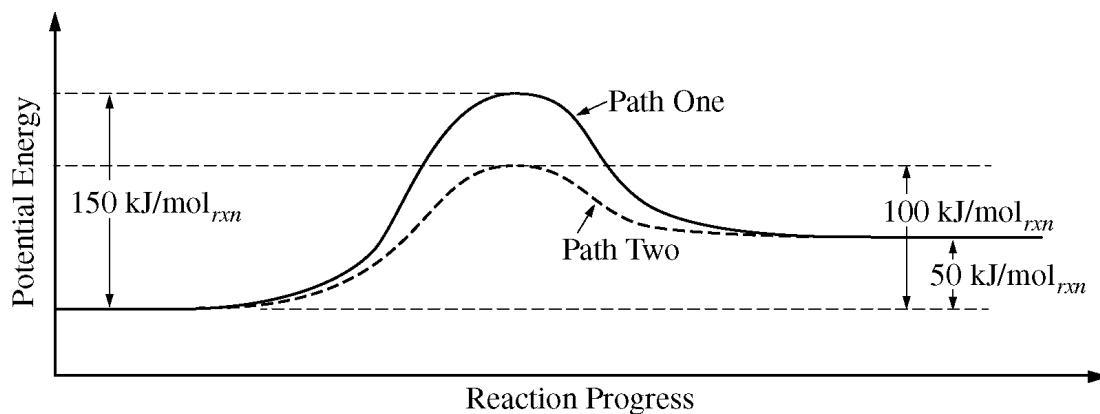
The diagram above illustrates how the reaction $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$ occurs on the surface of $\text{Ru}(s)$. The rate of this reaction is determined by the amount of energy required to break the bond in N_2 . This bond is weakened when N_2 is adsorbed on $\text{Ru}(s)$. Based on this information, which of the following provides the best reason for the use of $\text{Ru}(s)$ for the synthesis of NH_3 ?

- (A) It promotes the proper orientation of N and H atoms to form new N – H bonds.
- (B) It provides a reaction path with a lower activation energy.
- (C) It forms a bond between Ru and N that is stronger than the bond in N_2 .
- (D) It decreases the frequency of collisions between $\text{N}_2(g)$ and $\text{H}_2(g)$.

The following questions relate to the below information.

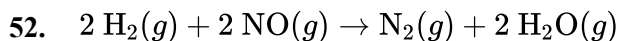


The equation above represents the decomposition of a compound XY_2 . The diagram below shows two reaction profiles (path one and path two) for the decomposition of XY_2 .

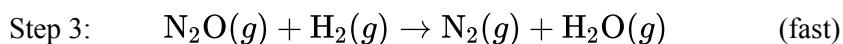
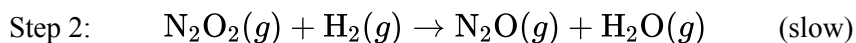


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51. Which of the following most likely accounts for the difference between reaction path one and reaction path two?
- (A) A higher temperature in path one
 - (B) A higher temperature in path two
 - (C) The presence of a catalyst in path one
 - (D) The presence of a catalyst in path two
-



For the chemical reaction represented above, the following mechanism is proposed.

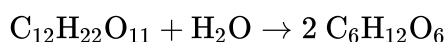
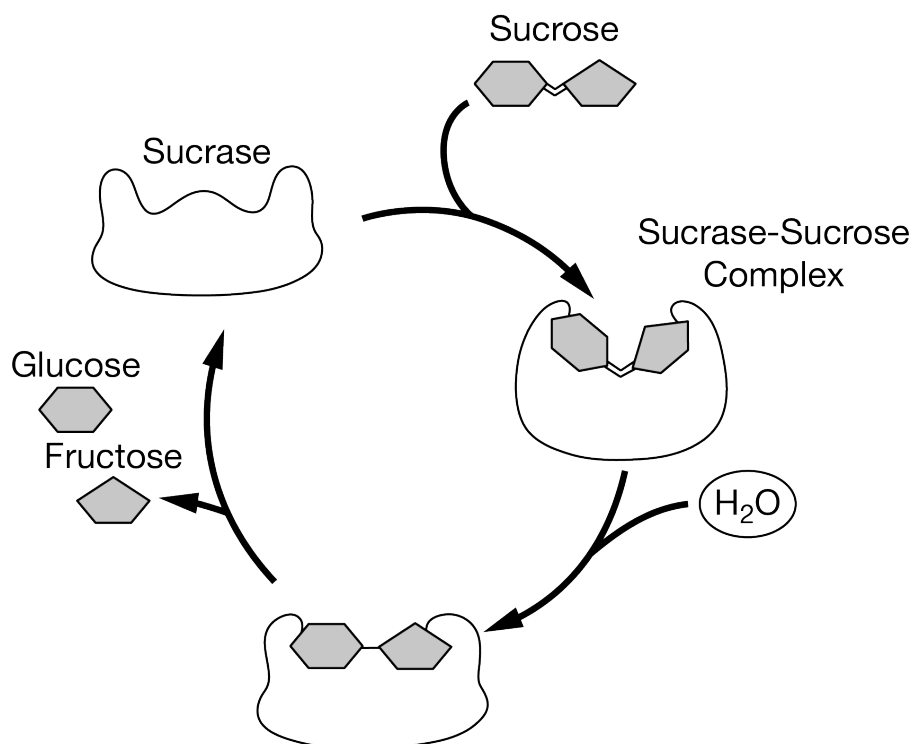


Which of the following initial rate law expressions is consistent with this proposed mechanism?

- (A) $\text{Rate} = k[\text{H}_2]^2[\text{NO}]^2$
- (B) $\text{Rate} = k[\text{NO}]^2$
- (C) $\text{Rate} = k[\text{H}_2][\text{NO}]^2$
- (D) $\text{Rate} = k[\text{H}_2][\text{N}_2\text{O}]^2$

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53.



The hydrolysis of sucrose is represented by the chemical equation above. This reaction is extremely slow in aqueous solution. However, when sucrase is added as shown in the diagram, the rate of the reaction is about 6,000,000 times faster. Based on this information, which of the following best explains the large increase in the rate of hydrolysis that occurs with the addition of sucrase?

- (A) The reaction proceeds through a different reaction path with a lower activation energy in which the sucrase-sucrose complex is formed as an intermediate.
- (B) The addition of the sucrase reduces the number of effective collisions between the water and sucrose molecules.
- (C) The reaction proceeds through a different reaction path that is independent of temperature and concentration.
- (D) The addition of the sucrase decreases the number of sucrose molecules that have the minimum energy required to overcome the activation energy barrier.

54. The role of a catalyst in a chemical reaction is to

- (A) decrease the amount of reactants that must be used
- (B) lower the activation energy for the reaction
- (C) supply the activation energy required for the reaction to proceed
- (D) increase the amounts of products formed at equilibrium
- (E) increase the entropy change for the reaction

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55. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, X and Z, are shown in the following table.

Compound	X	Z
Diagram	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & :\text{O}: & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{O}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & & & \end{array} $	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{S}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $
Boiling Point	437 K	371 K

- (a) Based on VSEPR theory, what is the approximate value of the O — C — O bond angle in compound X?
- b) The O — H bond in compound X is shorter than the S — H bond in compound Z. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.

Compound X —

Compound Z —

- (d) Explain why the boiling point of compound X is higher than that of compound Z based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound Z is placed in an evacuated, rigid container at a temperature of 800. K and a pressure of 1.00 atm. Determine the final pressure of the gas when the temperature is decreased to 600. K.

Another carbon-containing compound is $\text{C}_2\text{H}_6\text{O}$, which reacts with H_2CrO_4 , as represented by the following equation.



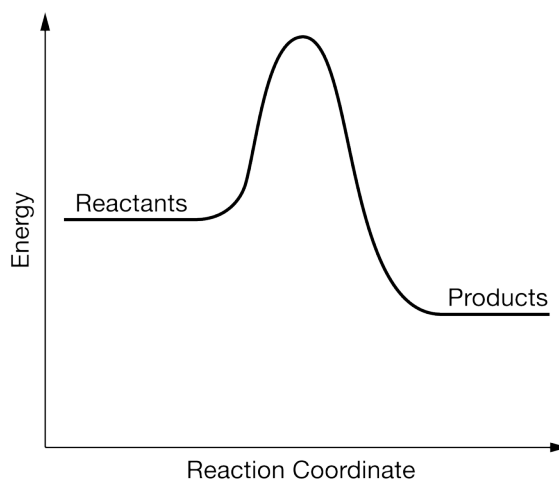
- (f) The student claims that C is reduced when $\text{C}_2\text{H}_6\text{O}$ is converted to $\text{C}_2\text{H}_4\text{O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

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Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 0.20 M/min . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

